

Strange series representations of $U_q(\mathfrak{su}(1, 1))$ and the $\text{AdS}_2 \leftrightarrow \text{dS}_2$ structure of the DSSYK bulk dual

Abstract

We examine the claim of arXiv:2512.10101 that the strange series representations of $U_q(\mathfrak{su}(1, 1))$ interpolate between the AdS_2 and dS_2 regions of the DSSYK bulk dual. *Proved:* the strange bilocal is the discrete bilocal times the chord-parity unitary Π implementing the normaliser $\mathbb{Z}_2$ -grading, and its equal-time kernel is the discrete kernel with one spectral argument reflected through the band centre — an exact finite- q identity; the normaliser fuses each strange representation with a discrete-series pair into one irreducible corepresentation ($a = \ell - \frac{1}{2}$); the momentum-space Casimir of the reduced quantum $\text{AdS}_{2,q}$ splits into two radial operators — the chord transfer matrix (wall) and a q -Liouville well with deficiency indices $(1, 1)$ whose self-adjoint extensions each carry one discrete-type *and* one strange-type eigenvalue tower, locked by an exact root-product rule, realising the fusion pairing at the Plancherel labels; and any normaliser-covariant AdS/dS criterion assigns the strange family and its discrete partners the same type. *Derived:* the classical limits (Liouville radial operator, Schwarzian density of states, Bessel- K wavefunctions); the strange towers merge into the AdS-edge discrete series as $q \rightarrow 1$ while their mirrors are expelled; an exact edge-connectivity transition in the complex bilocal weight, crossing precisely at the band-centre weight; and the identification of the $\alpha < -1$ q -Askey bound states of arXiv:2605.13956 with strange-series Casimir values. *Verdict:* the amplitude form of the claim holds conditional on the edge-dS identification (and fails under centre-dS); the geometric form is *refuted* — the $\text{AdS} \leftrightarrow \text{dS}$ bridge the framework supports is the grading Π , not a migration of the strange family, whose towers sit on the AdS side of the spectrum. *Open:* the extension selected by the rescaled quantum group, the grading unitary as a rescaled limit of the dual algebra, a rescaled Plancherel theory, the non-spherical bilocals, and a falsifiable conjecture for the announced spectral-triple metric. All numerical statements carry committed, bit-for-bit reproducible checks.

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1 Introduction and statement of the problem

1.1 The claim, verbatim

The object of this project is the following assertion of Schouten–Isachenkov [1] (arXiv:2512.10101v2; all quotations and equation numbers refer to v2; the full quotation ledger with locations is `lit/notes/2512.10101.md`):

“Lastly, we make remarks on the correlation function related to the strange series representation, which is argued to interpolate between the AdS and dS regions of our q -homogeneous space.” (abstract)

“such that the strange series has a non-zero amplitude between the lower part and upper part of the spectrum . It was proposed in [arXiv:2411.16922] and [arXiv:2505.08116] that zooming into the upper part of the spectrum, i.e. $\theta = |\ln(q^2)|k$, corresponds to de Sitter JT-gravity. If this proposal were proven to be correct, the strange series would therefore have a non-zero amplitude between the anti-de Sitter and de Sitter regions.” (§4.3, end)

“Moreover, supposing that dS space corresponds to the upper part of the spectrum, it was shown in Section 4.3 that the strange series gives non-zero amplitudes between AdS and dS regions, and thus in some sense interpolates between them. It is currently unknown what the correct physics interpretation of such an amplitude would be.” (§5, Discussion, item 1)

The authors' own epistemic labelling is therefore: *argued*, twice conditional — on the dS identification of the upper spectral edge, and on an interpretation of the amplitude that they declare unknown. The phrase “in some sense interpolates” is theirs; making the sense precise is the purpose of this section.

1.2 Setting and the paper's computation

Throughout, conventions are those of Section 2. The DSSYK spectral parameter is $\theta \in [0, \pi]$, $E(\theta) = 2 \cos \theta / \sqrt{1 - q^2}$ (Convention 2.1); the principal-series eigenvectors $|P^\theta\rangle$ and the chord states $|k\rangle$ are as in Conventions 2.3 and 2.7.

Definition 1.1 (edge regimes). For $k \geq 0$ fixed and $q \rightarrow 1^-$, the *lower-edge regime* is the family $\theta_q^L(k) := \pi - |\ln q^2| k$ and the *upper-edge regime* is $\theta_q^U(k) := |\ln q^2| k$.

Definition 1.2 (AdS region). The *AdS region* is the lower-edge regime. Justification (not a convention): in this regime the $q \rightarrow 1$ limit of DSSYK is the Schwarzian theory on AdS_2 [SI §4.2, eq. (4.17); standard DSSYK literature [2]], and every bulk-dual proposal on the market agrees on this identification.

Definition 1.3 (dS region — two inequivalent candidates). (a) *Edge-dS (dS-JT)*: the upper-edge regime $\theta_q^U(k)$ [arXiv:2411.16922 [3]; arXiv:2505.08116 [4]]. (b) *Centre-dS*: the middle of the band, $\theta = \pi/2$, $E = 0$ [arXiv:2310.16994 [5] and successors]. The claim of [1] is formulated with (a); its footnote records (b) as a rival. Any statement below that depends on the choice says so explicitly.

The paper's strange-series computation concerns the bilocal operators of Convention 2.7, $\mathcal{O}^\ell$ (discrete series) and $\mathcal{S}^\ell$ (strange series), $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$, through their equal-time overlap kernels

$$\mathcal{K}_q^\mathcal{O}(\theta_1, \theta_2) := \langle P^{\theta_1} | \mathcal{O}^\ell(0) | P^{\theta_2} \rangle = \frac{(q^{4\ell}; q^2)_\infty}{(q^{2\ell} e^{i(\pm\theta_1 \pm \theta_2)}; q^2)_\infty}, \quad \mathcal{K}_q^\mathcal{S}(\theta_1, \theta_2) := \langle P^{\theta_1} | \mathcal{S}^\ell(0) | P^{\theta_2} \rangle, \quad (1)$$

where the denominator is the product over the four sign choices [SI (4.15)–(4.16), (4.19)].

Lemma 1.4 (q -Mehler form; chord-side characterisation). [DERIVED] For $t \in \mathbb{C}$, $|t| < 1$,

$$\sum_{k \geq 0} \frac{t^k}{(q^2; q^2)_k} H_k(\cos \theta_1; q^2) H_k(\cos \theta_2; q^2) = \frac{(t^2; q^2)_\infty}{(t e^{i(\pm\theta_1 \pm \theta_2)}; q^2)_\infty}, \quad (2)$$

the q -Mehler formula for continuous q -Hermite polynomials (Gasper–Rahman; [2] eq. (2.28)). Consequently $\mathcal{K}_q^\mathcal{O}$ is (2) at $t = q^{2\ell}$, and $\mathcal{K}_q^\mathcal{S}$ is (2) at $t = -q^{2\ell}$: the strange bilocal inserts matter with alternating chord weights $(-1)^k q^{2k\ell}$, versus $q^{2k\ell}$ for the discrete bilocal. In particular $\mathcal{K}_q^\mathcal{S}(\theta_1, \theta_2) = (q^{4\ell}; q^2)_\infty / (-q^{2\ell} e^{i(\pm\theta_1 \pm \theta_2)}; q^2)_\infty$, which is SI (4.19) before its limit.

Proof. $\hat{\rho}_{st}|k\rangle = \frac{1}{2} q^{-1} q^{-2k} |k\rangle$ [SI (4.13)] gives $(2q\hat{\rho}_{st})^{-\ell - \frac{i\pi}{2\ln q}} |k\rangle = q^{2k(\ell + \frac{i\pi}{2\ln q})} |k\rangle = (-1)^k q^{2k\ell} |k\rangle$, since $q^{2k \cdot i\pi/(2\ln q)} = e^{i\pi k}$. Expanding both $|P^{\theta_i}\rangle$ in chord states (Convention 2.3) yields the left side of (2) with $t = \mp q^{2\ell}$; (2) itself is classical. $\square$

Lemma 1.5 (exact reflection identity). [DERIVED] For all $q \in (0, 1)$, $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$ and θ_1, θ_2 ,

$$\mathcal{K}_q^\mathcal{S}(\theta_1, \theta_2) = \mathcal{K}_q^\mathcal{O}(\theta_1, \pi - \theta_2) = \mathcal{K}_q^\mathcal{O}(\pi - \theta_1, \theta_2). \quad (3)$$

That is: at finite q , the strange-series equal-time kernel is the discrete-series kernel with one spectral argument reflected through the middle of the band ($E \rightarrow -E$).

Proof. $e^{\pm i(\pi-\theta_2)} = -e^{\mp i\theta_2}$, so the four denominator arguments $-q^{2\ell}e^{i(\pm\theta_1\pm\theta_2)}$ of $\mathcal{K}_q^{\mathcal{S}}(\theta_1, \theta_2)$ (Lemma 1.4) coincide, as a set, with the four arguments $q^{2\ell}e^{i(\pm\theta_1\pm(\pi-\theta_2))}$ of $\mathcal{K}_q^{\mathcal{O}}(\theta_1, \pi - \theta_2)$; the numerators are equal. The second equality holds because the kernel depends on (θ_1, θ_2) only through the four sums $\pm\theta_1 \pm \theta_2$, and reflecting both arguments returns the original kernel: $\mathcal{K}^{\mathcal{S}}$ and $\mathcal{K}^{\mathcal{O}}$ are each invariant under simultaneous reflection. $\square$

Proposition 1.6 (the paper's limit statements, re-derived). [DERIVED] Fix $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$ and $k_1, k_2 \geq 0$. As $q \rightarrow 1^-$:

- (i) Same edge, discrete (Schwarzian benchmark): $\mathcal{K}_q^{\mathcal{O}}(\theta_q^L(k_1), \theta_q^L(k_2)) \sim N(q) \frac{\Gamma(\ell \pm ik_1 \pm ik_2)}{\Gamma(2\ell)}$, where $\Gamma(\ell \pm ik_1 \pm ik_2)$ denotes the product over the four sign choices and $N(q) := (1 - q^2)^{2\ell-3}(q^2; q^2)_{\infty}^{-3}$ [SI (4.17)]; the q -gamma limit $\Gamma_{q^2} \rightarrow \Gamma$.
- (ii) Same edge, strange: the normalisation-free ratio vanishes, $\mathcal{K}_q^{\mathcal{S}}(\theta_q^L(k_1), \theta_q^L(k_2)) / \mathcal{K}_q^{\mathcal{O}}(\theta_q^L(k_1), \theta_q^L(k_2)) \rightarrow 0$ [this is the content of SI (4.19)].
- (iii) Opposite edges, strange: exactly, at every q , $\mathcal{K}_q^{\mathcal{S}}(\theta_q^L(k_1), \theta_q^U(k_2)) = \mathcal{K}_q^{\mathcal{O}}(\theta_q^L(k_1), \theta_q^L(k_2))$, hence with the same normalisation $N(q)$ as in (i) the limit is the Schwarzian $3j$ -symbol $\Gamma(\ell \pm ik_1 \pm ik_2) / \Gamma(2\ell)$ [this is SI (4.20), sharpened to an identity].
- (iv) Opposite edges, discrete: the ratio $\mathcal{K}_q^{\mathcal{O}}(\theta_q^L(k_1), \theta_q^U(k_2)) / \mathcal{K}_q^{\mathcal{O}}(\theta_q^L(k_1), \theta_q^L(k_2)) \rightarrow 0$: ordinary (discrete-series) matter does not connect the two edges.

Proof. $\theta_q^L(k)$ gives $e^{i\theta_q^L(k)} = -q^{2ik}$ and $\theta_q^U(k)$ gives $e^{i\theta_q^U(k)} = q^{-2ik}$. (iii) is then Lemma 1.5 evaluated at $\pi - \theta_q^U(k_2) = \theta_q^L(k_2)$ — the reflection is exact, no limit needed — combined with (i). For (i): each denominator factor is $(q^{2(\ell \pm ik_1 \pm ik_2)}; q^2)_{\infty}$, and $(q^{2x}; q^2)_{\infty} = (1 - q^2)^{1-x}(q^2; q^2)_{\infty} / \Gamma_{q^2}(x)$ [SI (A.3)] with $\Gamma_{q^2}(x) \rightarrow \Gamma(x)$ gives the stated asymptotics; the imaginary parts cancel in the accumulated exponent of $(1 - q^2)$. For (ii) and (iv): by Lemma 1.5 the two ratios are equal, so it suffices to prove (iv). By (i)'s factorisation, the ratio in (iv) is $\prod_{\pm\pm} (q^{2(\ell \pm ik_1 \pm ik_2)}; q^2)_{\infty} / \prod_{\pm\pm} (-q^{2(\ell \pm ik_1 \pm ik_2)}; q^2)_{\infty}$. Fix one of the four sign choices and set $x := \ell \pm ik_1 \pm ik_2$, so that

$$q^{2x+2n} = \rho_n e^{i\varphi_q}, \quad \rho_n := q^{2\ell+2n} \in (0, 1), \quad \varphi_q := 2(\pm k_1 \pm k_2) \ln q,$$

the phase φ_q being independent of n . Both infinite products converge absolutely, and $|1 + q^{2x+2n}| \geq 1 - \rho_n \geq 1 - q^{2\ell} > 0$ for every n , so each factor $r_x(q) := (q^{2x}; q^2)_{\infty} / (-q^{2x}; q^2)_{\infty}$ is a well-defined nonzero number with

$$|r_x(q)|^2 = \prod_{n \geq 0} \frac{|1 - \rho_n e^{i\varphi_q}|^2}{|1 + \rho_n e^{i\varphi_q}|^2} = \prod_{n \geq 0} \frac{1 - 2\rho_n \cos \varphi_q + \rho_n^2}{1 + 2\rho_n \cos \varphi_q + \rho_n^2}.$$

Assume q close enough to 1 that $2(k_1 + k_2)|\ln q| \leq \pi/3$ (for $k_1 = k_2 = 0$ no condition is needed); then $|\varphi_q| \leq \pi/3$, hence $\cos \varphi_q \geq \frac{1}{2}$, for all four sign choices. The per-factor ratio is decreasing in $\cos \varphi_q$ (its derivative in $c = \cos \varphi_q$ is $-4\rho_n(1 + \rho_n^2)$ divided by the squared denominator, hence negative), so for every n

$$\frac{1 - 2\rho_n \cos \varphi_q + \rho_n^2}{1 + 2\rho_n \cos \varphi_q + \rho_n^2} \leq \frac{1 - \rho_n + \rho_n^2}{1 + \rho_n + \rho_n^2} =: g(\rho_n) < 1.$$

The function g is strictly decreasing on $(0, 1)$: the numerator of g' is $(2\rho - 1)(1 + \rho + \rho^2) - (1 - \rho + \rho^2)(2\rho + 1) = 2\rho^2 - 2 < 0$. Hence $g(\rho_n) \leq g(\frac{1}{2}) = \frac{3}{7}$ whenever $\rho_n \geq \frac{1}{2}$, which happens exactly for $n \leq N_{\ell}(q) := \ln 2 / (2|\ln q|) - \ell$; the number of such lattice sites tends to ∞ as $q \rightarrow 1^-$. Every remaining factor is ≤ 1 . Therefore

$$|r_x(q)|^2 \leq \left(\frac{3}{7}\right)^{\lfloor N_{\ell}(q) \rfloor + 1} \xrightarrow{q \rightarrow 1^-} 0 \quad (\text{for } q \text{ with } N_{\ell}(q) \geq 0),$$

so each of the four factors tends to 0, and with it the ratio in (iv); (ii) then follows from the equality of the two ratios established above. The quantitative bound is verified numerically in `src/check_claim_kernels.py`, check (K5) (output `results/check_claim_kernels.out`): at $q = 0.9, 0.99, 0.999$ (with $\ell = 1, k_1 = 0.8, k_2 = 1.3$) the smallness hypothesis holds ($2(k_1 + k_2)|\ln q| \leq 0.443$), every per-factor ratio with $\rho_n \geq \frac{1}{2}$ is $\leq \frac{3}{7}$, and the worst measured $|r_x(q)|^2$ over the four sign choices is 2.2×10^{-7} against the bound 7.9×10^{-2} at $q = 0.9$, 8.3×10^{-103} against 3.1×10^{-13} at $q = 0.99$, and 3.2×10^{-1066} against 4.8×10^{-128} at $q = 0.999$ (the bound is far from tight, as expected of a counting bound); the full ratio falls to 10^{-21419} by $q = 0.9999$, the renormalised mixed-edge strange kernel agrees with $\Gamma(\ell \pm ik_1 \pm ik_2)/\Gamma(2\ell)$ to 4.7×10^{-4} relative at $q = 0.9999$, and the reflection identity holds to better than 9×10^{-37} relative at all sampled q , with further scans in issue #10. *Note*: SI's printed (4.19) writes the strange shift as $i\pi/\ln q$ (the full period of Γ_{q^2} , under which the corresponding q -gamma ratio would *not* vanish); consistency with their own (4.18) and §4.3 text requires $i\pi/(2\ln q)$; see `lit/notes/2512.10101.md` §7. *Tag bookkeeping* (issue #18): the vanishing in (ii)/(iv) is proved in full by the product bound above; the proposition as a whole keeps the tag [DERIVED] because (i) and (iii) rest on the q -gamma limit $\Gamma_{q^2} \rightarrow \Gamma$ and on Lemma 1.4, which are complete calculations but not independently re-proved here. $\square$

1.3 Formalisations of the claim

The informal claim admits inequivalent formalisations. We state all that we will use, from weakest to strongest, and identify which one this project attacks.

Claim 1.7 (C-amp: non-vanishing $\text{AdS} \leftrightarrow \text{dS}$ amplitude; SI's literal claim). *Assume Definition 1.3(a) (edge-dS). Then the strange-series bilocals $\mathcal{S}^\ell$, $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$, have equal-time overlap kernels between the AdS region and the dS region with finite non-zero $q \rightarrow 1$ limit (after the normalisation $N(q)$ that renders the discrete-series AdS kernel finite), given by the Schwarzian $3j$ -symbol; their $\text{AdS} \leftrightarrow \text{AdS}$ kernels vanish in the same normalisation; and for the discrete-series bilocals the situation is reversed.* Status: [DERIVED] — this is Proposition 1.6 (ii)–(iv), conditional only on the meaning of “dS”.

Claim 1.8 (C-geo: geometric interpolation; the open claim this project attacks). *There exists a radial Casimir problem on the quantum homogeneous space carrying the DSSYK dynamics (issues #5–#7) — a second-order q -difference operator in an invariant coordinate, with an effective potential depending on the representation label — together with a criterion, fixed in advance (issue #8), for when such a radial problem is of AdS_2 type and when of dS_2 type; and under this criterion the strange series realises the dS_2 type, or crosses from one type to the other as its label varies, while the representations with classical analogues realise the AdS_2 type.* Status: [CONJECTURED] — this is the precise reading of “genuinely interpolate”; nothing in [1] proves or refutes it, and the authors call the interpretation of the amplitude “currently unknown”. Resolution (issues #7–#8): *refuted* — see Theorem 6.5, whose classes (I)–(III) exhaust the readings of this claim.

Remark 1.9 (what would settle the question). Claim 1.7 is settled (Proposition 1.6), *given* its conditionality; what remains open in it is exactly its hypothesis, Definition 1.3(a). Claim 1.8 is open and falsifiable: computing the radial Casimir operator on the space fixed in issue #5, restricted to the strange series (issues #6–#7), and evaluating the issue-#8 criterion on it, either exhibits the $\text{AdS} \rightarrow \text{dS}$ crossing (proving Claim 1.8), or exhibits an obstruction (refuting it). A specialist can check either outcome against Sections 3–6.

Remark 1.10 (dS-identification dependence). [ARGUED] Claim 1.7 is *not* robust against replacing Definition 1.3(a) by (b). At $\theta_2 = \pi/2$ the four denominator arguments of $\mathcal{K}_q^S(\theta_q^L(k_1), \pi/2)$

tend to $\pm i$, and the standard asymptotics $\ln(u; q^2)_\infty \sim -\text{Li}_2(u)/|\ln q^2|$ gives

$$\mathcal{K}_q^S(\theta_q^L(k_1), \pi/2) / \mathcal{K}_q^O(\theta_q^L(k_1), \theta_q^L(k_2)) \sim \exp\left(-\frac{3\pi^2}{4|\ln q^2|}\right) \rightarrow 0$$

(using $\text{Li}_2(i) + \text{Li}_2(-i) = -\pi^2/24$, $\text{Li}_2(1) = \pi^2/6$): the strange amplitude between the AdS edge and the band centre is exponentially small in the same normalisation in which the edge-to-edge amplitude is finite. Numerical check (issue #10 scans pending): the ratio is 5.6×10^{-11} at $q = 0.9$, 1.8×10^{-153} at $q = 0.99$, 1.3×10^{-313} at $q = 0.995$; the measured rate $|\ln q^2| \cdot \ln(\text{ratio})$ is $-4.97, -7.07, -7.22$ against the predicted $-3\pi^2/4 = -7.40$ (subleading corrections are $O(|\ln q^2| \ln |\ln q^2|)$ in the exponent). So if the centre-dS identification (b) is the correct one, the strange series does *not* connect AdS to dS in the amplitude sense — the claim's truth value depends on which dS proposal one adopts. This criterion-sensitivity is exactly the situation flagged in issue #8 (“if the criterion is ambiguous, that is itself the finding”); the quantitative treatment belongs to #8/#11.

Remark 1.11 (discreteness of the strange family). “Interpolate” suggests a continuous deformation. Three parameters could supply it: (1) the abstract strange label $a > 0$, continuous in the classification (Convention 2.5) but *discrete* ($a \in \frac{1}{2}\mathbb{Z}_{>0}$) in the Plancherel decomposition of the regular representation (Convention 2.6) — and the bilocal construction of SI uses $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$ because integer powers of the spherical element are what exists in the coordinate algebra (their footnote on the universal cover for continuous weights applies here too); (2) the spectral parameter θ (continuous; this is the parameter in Claim 1.7); (3) the radial coordinate of the homogeneous space (continuous q -lattice; this is the parameter in Claim 1.8). Any proof or refutation of Claim 1.8 must say against which parameter the “crossing” happens; a formalisation in which no continuous parameter crosses anything is not an interpolation statement but a two-point statement, and must be reported as such.

Remark 1.12 (degenerate readings, excluded). Two readings are excluded as contentless. (a) “The strange series are the negative- $\tilde{\Omega}$ counterparts of the discrete series” is true by construction (Convention 2.5); without an independent geometric criterion it says nothing about AdS or dS. (b) Given Definition 1.3(a) and Proposition 1.6, calling the non-zero mixed-edge kernel “interpolation” is definitionally true; the content resides in the hypothesis (a) and in the *selectivity* of the property — among the bilocal families computed so far (discrete, strange) exactly the strange one connects the edges. Whether the selectivity survives the inclusion of principal-series bilocals is, to our knowledge, not computed anywhere; we record it as an open sub-question for issue #6.

1.4 The normaliser origin of the reflection identity

Lemma 1.5 and the normaliser structure of Convention 2.4 are two occurrences of one mechanism. This subsection makes the connection precise: the extra factor distinguishing the strange bilocal from the discrete bilocal is a self-adjoint unitary Π that anticommutes with the transfer matrix — the chord-sector shadow of the $\mathbb{Z}_2$ -grading of the dual von Neumann algebra of [6] — and at the unrescaled level the strange and discrete series are *fused into single irreducible corepresentations* of $\text{SU}_q(1, 1) \rtimes \mathbb{Z}_2$. The structural consequence (Theorem 1.21) is an obstruction relevant to issue #8.

Lemma 1.13 (the grading unitary on the chord sector). *[PROVED] Let $q \in (0, 1)$, let H_{ch} be the closure of the span of the chord states $\{|k\rangle\}_{k \in \mathbb{Z}_{\geq 0}}$ (an orthonormal basis, Convention 2.7, boundary conditions $\eta = \xi = 1$), and let $2q\hat{\rho}_{st}$ be the positive self-adjoint diagonal operator $2q\hat{\rho}_{st}|k\rangle = q^{-2k}|k\rangle$ [SI (4.13)]. Define, by the Borel functional calculus with the real branch of $\ln$ on $(0, \infty)$,*

$$\Pi := (2q\hat{\rho}_{st})^{-\frac{i\pi}{2\ln q}}, \quad z^{-\frac{i\pi}{2\ln q}} := \exp\left(-\frac{i\pi \ln z}{2\ln q}\right). \quad (4)$$

Then:

- (i) $\Pi|k\rangle = (-1)^k|k\rangle$; consequently $\Pi = \Pi^* = \Pi^{-1}$, $\Pi^2 = 1$: Π is the chord-parity involution, the nontrivial character of the spectral lattice $q^{-2\mathbb{Z}_{\geq 0}}$ of $2q\hat{\rho}_{st}$.
- (ii) $\Pi\tilde{\Omega}\Pi = -\tilde{\Omega}$ on H_{ch} , hence $\Pi T\Pi = -T$ and $\Pi e^{\tau T}\Pi = e^{-\tau T}$ for every $\tau \in \mathbb{C}$ ($T = 2\tilde{\Omega}/\sqrt{1-q^2}$, Convention 2.3).
- (iii) Coefficientwise on principal eigenvectors: with $v_k(\theta) := H_k(\cos\theta; q^2)/\sqrt{(q^2; q^2)_k}$ (Convention 2.3), $(-1)^k v_k(\theta) = v_k(\pi - \theta)$ for all $k \geq 0$ and $\theta \in [0, \pi]$; that is, $\Pi|P^\theta\rangle = |P^{\pi-\theta}\rangle$.

Proof. (i) On the eigenvalue $z = q^{-2k}$, $\ln z = -2k \ln q$, so $z^{-i\pi/(2\ln q)} = \exp(i\pi k) = (-1)^k$. The function takes values in $\{\pm 1\} \subset \mathbb{R}$, so Π is a self-adjoint unitary with $\Pi^2 = 1$. (ii) $\tilde{\Omega}$ is bounded on H_{ch} : writing $\tilde{\Omega} = A + A^*$ with $A|k\rangle = \frac{1}{2}\sqrt{1-q^{2k}}|k-1\rangle$, $\|A\| \leq \frac{1}{2}$ gives $\|\tilde{\Omega}\| \leq 1$. On basis vectors, $\Pi\tilde{\Omega}\Pi|k\rangle = (-1)^k(\frac{1}{2}\sqrt{1-q^{2k+2}}(-1)^{k+1}|k+1\rangle + \frac{1}{2}\sqrt{1-q^{2k}}(-1)^{k-1}|k-1\rangle) = -\tilde{\Omega}|k\rangle$, and both sides are bounded, so the identity extends to H_{ch} by linearity and continuity. The exponential statement follows term by term from the (norm-convergent, since T is bounded) power series of $e^{\tau T}$: $\Pi T^n \Pi = (-T)^n$. (iii) $H_k(-x; q^2) = (-1)^k H_k(x; q^2)$ by induction on k from the recurrence $H_{k+1}(x) = 2xH_k(x) - (1-q^{2k})H_{k-1}(x)$: it holds for $H_0 = 1$ and $H_1 = 2x$, and if it holds at $k-1, k$ then $H_{k+1}(-x) = -2xH_k(-x) - (1-q^{2k})H_{k-1}(-x) = (-1)^{k+1}(2xH_k(x) - (1-q^{2k})H_{k-1}(x)) = (-1)^{k+1}H_{k+1}(x)$. Apply this at $x = \cos\theta$, $-x = \cos(\pi - \theta)$. (Numerical confirmation to 10^{-59} or better: checks (G1)–(G3).) $\square$

Proposition 1.14 (factorisation of the strange bilocal; the exact content of Lemma 1.5). [PROVED] Let $q \in (0, 1)$ and $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$, with $\mathcal{O}^\ell, \mathcal{S}^\ell$ as in Convention 2.7. Then:

- (a) (factorisation) As bounded operators on H_{ch} ,

$$\mathcal{S}^\ell(0) = \mathcal{O}^\ell(0)\Pi = \Pi\mathcal{O}^\ell(0), \quad \mathcal{S}^\ell(\tau) = \mathcal{O}^\ell(\tau)e^{\tau T}\Pi e^{-\tau T} = \mathcal{O}^\ell(\tau)\Pi e^{-2\tau T} \quad (\tau \in \mathbb{C}). \quad (5)$$

- (b) (the conjugation reading is false) $\Pi\mathcal{O}^\ell(0)\Pi^{-1} = \mathcal{O}^\ell(0)$: conjugation by Π fixes every function of $\hat{\rho}_{st}$, and $\Pi\mathcal{O}^\ell(0)\Pi^{-1} \neq \mathcal{S}^\ell(0)$ (they differ by the factor $(-1)^k$ on every chord state with k odd). The strange bilocal is not the discrete bilocal conjugated by a normaliser element; it is the discrete bilocal multiplied by the grading unitary Π .

- (c) (kernel identity through Π) For $\theta_1, \theta_2 \in [0, \pi]$ the chord sums

$$\sum_{k \geq 0} v_k(\theta_1) (-1)^k q^{2\ell k} v_k(\theta_2) = \sum_{k \geq 0} v_k(\theta_1) q^{2\ell k} v_k(\pi - \theta_2) \quad (6)$$

converge absolutely and are equal; i.e. $\langle P^{\theta_1} | \mathcal{S}^\ell(0) | P^{\theta_2} \rangle = \langle P^{\theta_1} | \mathcal{O}^\ell(0) | P^{\pi-\theta_2} \rangle$ termwise. Combined with Lemma 1.4 this reproduces Lemma 1.5; the identity (6) itself is proved here without the q -Mehler formula.

Proof. (a) The Borel functional calculus of the positive self-adjoint operator $2q\hat{\rho}_{st}$ is multiplicative, and $z^{-\ell - \frac{i\pi}{2\ln q}} = z^{-\ell} \cdot z^{-\frac{i\pi}{2\ln q}}$ pointwise on $(0, \infty)$; evaluating on the spectrum, $\mathcal{S}^\ell(0)|k\rangle = (-1)^k q^{2\ell k}|k\rangle = \mathcal{O}^\ell(0)\Pi|k\rangle$. All three operators are diagonal in the chord basis, hence commute pairwise. The τ -dependent statement follows by conjugating with $e^{\tau T}$ (bounded for all $\tau \in \mathbb{C}$ since T is bounded) and using $\Pi e^{-\tau T} = e^{\tau T}\Pi$ (Lemma 1.13(ii)). (b) Diagonal operators commute, so $\Pi\mathcal{O}^\ell(0)\Pi^{-1} = \mathcal{O}^\ell(0)$; on $|k\rangle$ with k odd, $\mathcal{O}^\ell(0)|k\rangle = q^{2\ell k}|k\rangle \neq -q^{2\ell k}|k\rangle = \mathcal{S}^\ell(0)|k\rangle$. (c) Absolute convergence: by the explicit expansion $H_k(\cos\theta; q^2) = \sum_{j=0}^k \begin{bmatrix} k \\ j \end{bmatrix}_{q^2} e^{i(k-2j)\theta}$ [KLS [7] §14.26] and $\begin{bmatrix} k \\ j \end{bmatrix}_{q^2} \leq (q^2; q^2)_\infty^{-2}$, one has $|H_k(\cos\theta; q^2)| \leq (k+1)(q^2; q^2)_\infty^{-2}$, so both series in (6) are dominated by $\sum_k (k+1)^2 (q^2; q^2)_\infty^{-5} q^{2\ell k} < \infty$. Termwise equality is Lemma 1.13(iii) applied to $v_k(\theta_2)$. (Checks (G4)–(G5): factorisation and conjugation triviality exact; kernel sums agree to relative 3×10^{-60} and match the closed form of $\mathcal{K}_q^\mathcal{O}(\theta_1, \pi - \theta_2)$ to relative 4×10^{-60} .) $\square$

Proposition 1.15 (the $\mathbb{Z}_2$ -grading of the dual algebra; GKK). *[PROVED] (in [6]; transcription, with the dictionary of Convention 2.8; numerically verified on truncations, checks (G6).) Let (M, Δ) be the von Neumann algebraic quantum group $\mathrm{SU}_q(1, 1) \rtimes \mathbb{Z}_2$ and $\hat{M}$ its dual, acting on the GNS space $\mathcal{K}$ of the Haar weight. Then:*

- (i) *With $\mathcal{K}_\pm := \overline{\mathrm{span}}\{f_{m,p,t} : \mathrm{sgn}(pt) = \pm\}$ and $\hat{M}_\pm := \{x \in \hat{M} : x\mathcal{K}_\pm \subseteq \mathcal{K}_{\pm(\mp)}\}$ one has $\hat{M} = \hat{M}_+ \oplus \hat{M}_-$, and the Casimir satisfies the graded commutation relations $x\Omega_{\mathrm{GKK}} \subseteq \Omega_{\mathrm{GKK}}x$ for $x \in \hat{M}_+$ and $y\Omega_{\mathrm{GKK}} \subseteq -\Omega_{\mathrm{GKK}}y$ for $y \in \hat{M}_-$ [GKK Def. (defKpmandMpm), Prop. (decompMintoM+andM-andgradedcommutation)]. Since $\tilde{\Omega} = -\Omega_{\mathrm{GKK}}$ (Convention 2.8), the same relations hold verbatim for $\tilde{\Omega}$.*
- (ii) *$\hat{M}$ is generated by K, E, U_0^{+-}, U_0^{-+} , where the two extra generators lie in the odd part $\hat{M}_-$ and are necessary: E and K alone do not suffice, precisely because they commute with the Casimir while $\hat{M}$ does not [GKK Thm. (generatorsforhatM), Prop. (QppngeneratedualM), Lemma (polardecompositionQppn)].*
- (iii) *On the eigenvectors $e_m^{\epsilon,\eta}(p, x)$ of Ω_{GKK} for the eigenvalue $\epsilon\eta x$ in the sector (p, m, ϵ, η) , the generators act by [GKK Lemma (actionsofgeneratorsoneigvetsOm), eq. (introprincipalseries)]*

$$\begin{aligned} K e_m^{\epsilon,\eta}(p, x) &= p^{\frac{1}{2}} q^m e_m^{\epsilon,\eta}(p, x), & (q^{-1} - q) E e_m^{\epsilon,\eta}(p, x) &= q^{-m-\frac{1}{2}} p^{-\frac{1}{2}} \sqrt{1 + 2\epsilon\eta x q^{2m+1}p + q^{4m+2}p^2} e_{m+}^{\epsilon,\eta} \\ U_0^{+-} e_m^{\epsilon,\eta}(p, x) &= \eta(-1)^{v(p)} e_m^{\epsilon,-\eta}(p, x), & U_0^{-+} e_m^{\epsilon,\eta}(p, x) &= \epsilon\eta^{\chi(p)}(-1)^m e_m^{-\epsilon,\eta}(p, x), \end{aligned} \quad (7)$$

where $\chi(p) = \log_q p$ and $v(p) = k$ for $p = q^{2k}$ or $p = q^{2k-1}$. In particular both U_0^{+-} and U_0^{-+} flip the sign $\epsilon\eta$, hence map Ω_{GKK} -eigenvectors for λ to eigenvectors for $-\lambda$; they commute with K ; on the principal corepresentation space $\mathcal{L}_{p,x} = \bigoplus_{\epsilon,\eta} \ell_{\epsilon,\eta}^2(p, x)$ they are unitary and satisfy $(U_0^{+-})^2 = (U_0^{-+})^2 = -1$; and conjugation by them maps the E -action to the $(\epsilon\eta \mapsto -\epsilon\eta)$ -twisted action, with an extra factor -1 for U_0^{-+} (the $(-1)^m$ in (7) — the same parity twist as Π , in the K -weight index). (Checks (G6a)–(G6d): all verified to 1.5×10^{-59} or exactly, at $(q, \chi(p)) = (0.6, 0)$ and $(0.8, 1)$.)

Theorem 1.16 (fusion of strange and discrete series in the normaliser quantum group). *[PROVED] (in [6], Prop. (discretesubrepresinregrepr) and eq. (sudecompdiscretesubrepresinregrepr); label computation and transcription ours, numerically verified, checks (G7).) Let $q \in (0, 1)$, $p \in q^{\mathbb{Z}}$, and $x = \mu(\lambda)$ with $\lambda \in -q^{2\mathbb{Z}+1}p \cup q^{2\mathbb{Z}+1}p$, $|\lambda| > 1$; define $j, l \in \mathbb{Z}$ by $|\lambda| = q^{1-2j}p^{-1} = q^{1+2l}p$ (equivalently $l = -j - \chi(p)$) and*

$$\ell := j + \frac{1}{2}\chi(p) \in \frac{1}{2}\mathbb{Z}, \quad q^{2\ell-1} = |\lambda|^{-1}, \quad |\lambda| > 1 \iff \ell \geq 1. \quad (8)$$

Then (p, x) determines an irreducible discrete-series corepresentation $W_{p,x}$ of $\mathrm{SU}_q(1, 1) \rtimes \mathbb{Z}_2$ inside the left regular corepresentation, and as a $U_q(\mathfrak{su}(1, 1))$ -module

$$W_{p,x}|_{U_q(\mathfrak{su}(1,1))} \cong \pi_{\ell-\frac{1}{2}, \epsilon(p)}^S \oplus D_\ell^- \oplus D_\ell^+, \quad \epsilon(p) = \frac{1}{2}\chi(p) \bmod 1, \quad (9)$$

with $\tilde{\Omega} = -\mu(q^{2\ell-1})$ on the strange summand and $\tilde{\Omega} = +\mu(q^{2\ell-1})$ on both discrete summands. Moreover:

- (a) (exhaustiveness) $W \cong \bigoplus_{p \in q^{\mathbb{Z}}} (\int_{-1}^1 W_{p,x} dx \oplus \bigoplus_{x \in \sigma_d(\Omega_p)} W_{p,x})$ [GKK Thm. (decompositionofleftregularcorep)], and the corepresentations in the continuous part restrict to direct sums of two principal series [GKK Rmk. (propintroprincipalseriescorepresentation)]. Hence every occurrence of a strange series in the Plancherel decomposition of $L_q^2(\mathrm{SU}(1, 1) \rtimes \mathbb{Z}_2)$ lies inside a single irreducible corepresentation (9) whose other $U_q(\mathfrak{su}(1, 1))$ -constituents are $D_\ell^\pm$ with $a = \ell - \frac{1}{2}$ — exactly the label substitution $\ell \rightarrow \ell + \frac{i\pi}{2\ln q}$, $a = \ell - \frac{1}{2}$, by which [1] generate the strange series (Convention 2.5).

- (b) (the odd generators realise the exchange) *In the case $x > 0$ the strange summand is the block $(\epsilon, \eta) = (+, +)$ (all $m \in \mathbb{Z}$), and U_0^{+-} maps the D_ℓ^+ summand (block $(+, -)$, $m \geq j$) isometrically into it, while U_0^{-+} maps the D_ℓ^- summand (block $(-, +)$, $m \leq l$) isometrically into it; restricted to $W_{p,x}$ they are partial isometries. The cases $x < 0$ are identical with the roles of the blocks permuted [GKK Prop. (discretesubrepresinregrepr), cases (ii)–(iii)].*

Proof of the label computation and the transcription. GKK state the decomposition (in their conventions, with their labels) as $\pi_{\frac{1}{2}(\chi(p)-1)+j, \epsilon(p)}^S \oplus D_{-\frac{1}{2}\chi(p)-l}^- \oplus D_{\frac{1}{2}\chi(p)+j}^+$. From $l = -j - \chi(p)$ (which follows from $q^{1-2j}p^{-1} = q^{1+2l}p$), the D^- label is $-\frac{1}{2}\chi(p) - l = j + \frac{1}{2}\chi(p) = \ell$, equal to the D^+ label, and the strange label is $a = \frac{1}{2}(\chi(p) - 1) + j = \ell - \frac{1}{2}$. The Casimir values: on the strange block $\Omega_{\text{GKK}} = +|x| = \mu(q^{2a})$ with $q^{2a} = q^{2j-1}p = |\lambda|^{-1}$, and on the discrete blocks $\Omega_{\text{GKK}} = -|x| = -\mu(q^{1-2\ell})$; translation by $\tilde{\Omega} = -\Omega_{\text{GKK}}$ gives the values stated. The identification of the blocks with the printed matrix elements [GKK eqs. (strangeserrep), (pos-discrserrep), (negdiscrserrep)] under the index maps $n + \epsilon(p) = m + \frac{1}{2}\chi(p)$ (strange block), $n = m - j$ (D^+ block) and $n = l - m$ (D^- block) is a direct computation with the factorisation $1 + 2\epsilon\eta x q^{2m+1}p + q^{4m+2}p^2 = (1 + \epsilon\eta q^{2m+1}p\lambda)(1 + \epsilon\eta q^{2m+1}p\lambda^{-1})$; it is carried out numerically to 4×10^{-61} relative for all three blocks at $(q, \chi(p), j) = (0.6, 0, 2), (0.8, 1, 1), (0.7, 2, 3)$ (checks (G7), including the vanishing of the lower-edge factor $1 - q^{2m+2-2j}$ at $m = j - 1$, which is what makes the one-sided blocks invariant). Part (b) is (7) restricted to $W_{p,x}$, where $e_m^{\epsilon, \eta}(p, x)$, resp. $e_m^{-\epsilon, \eta}(p, x)$, is zero whenever the corresponding eigenvalue is not in the discrete spectrum of the target sector [GKK, definition of $e_m^{\epsilon, \eta}(p, x)$ in Sec. (ssec:discreteseries)]. $\square$

Remark 1.17 (GKK’s own summary). [6] state the phenomenon verbatim in their introduction: “However, in the quantum group case the discrete series are no longer split up into a positive discrete series and a negative discrete series.” Theorem 1.16 is the precise form: the normaliser fuses D_ℓ^+ , D_ℓ^- and $\pi_{\ell-1/2}^S$ into one irreducible corepresentation. Classically the normaliser fuses only $D_\ell^+ \oplus D_\ell^-$ (an L-packet, as [1] §2.3 note); the adjunction of the strange constituent is the quantum novelty.

Lemma 1.18 (essential uniqueness of Π). [PROVED] *Let V be a unitary operator on H_{ch} with $V\tilde{\Omega} = -\tilde{\Omega}V$. Then $V = \Pi f(\tilde{\Omega})$ for some Borel function f with $|f| = 1$ almost everywhere with respect to the spectral measure of $\tilde{\Omega}$ at $|0\rangle$. In particular Π is the unique self-adjoint-unitary implementation of the grading on the chord sector up to a spectral phase.*

Proof. ΠV commutes with $\tilde{\Omega}$: $\Pi V\tilde{\Omega} = -\Pi\tilde{\Omega}V = \tilde{\Omega}\Pi V$ by Lemma 1.13(ii). By Proposition 3.6, $\tilde{\Omega}$ on H_{ch} is unitarily equivalent to multiplication by x on $L^2([-1, 1], \nu)$ with ν the (absolutely continuous) q^2 -Hermite orthogonality measure and cyclic vector 1. A bounded operator B on $L^2(\nu)$ commuting with multiplication by x commutes with multiplication by any polynomial, hence $B(g) = B(1) \cdot g$ first for polynomials g and then, by continuity of B and density, for all $g \in L^2(\nu) \cap L^\infty(\nu)$; $h := B(1)$ satisfies $\|hg\|_2 \leq \|B\| \|g\|_2$ for all such g , which forces $|h| \leq \|B\|$ a.e. (apply it to indicators of the sets $\{|h| > \|B\| + \epsilon\}$), so $B = M_h$ with $h \in L^\infty(\nu)$. Thus $\Pi V = f(\tilde{\Omega})$; unitarity of ΠV gives $|f| = 1$ ν -a.e., and $V = \Pi f(\tilde{\Omega})$ since $\Pi^2 = 1$. $\square$

Remark 1.19 (what is *not* established: Π as a rescaled limit). [GAP: ISSUE #16] Lemma 1.13 and Proposition 1.15 identify Π with the normaliser grading *structurally*: Π anticommutes with $\tilde{\Omega}$ on the chord sector exactly as the odd part $\hat{M}_-$ anticommutes with $\tilde{\Omega}$ on the GNS space, and by Lemma 1.18 it is the essentially unique such unitary. What is not established is the *limit* identification: whether Π is the $r \rightarrow \infty$ rescaled limit (SI (4.2)) of a concrete element of $\hat{M}_-$ (for instance of U_0^{-+} in the appropriate sector, whose $(-1)^m$ factor in (7) is the same parity twist). SI’s rescaling is performed on states and on the spherical element, not on $\hat{M}_-$; issue #16 records this gap. No statement below depends on it.

Definition 1.20 (normaliser-covariant assignment). Let $\mathcal{R}$ be the set of irreducible admissible unitary $*$ -representations of $U_q(\mathfrak{su}(1,1))$ (Convention 2.5). An *assignment* is a map C from $\mathcal{R}$ to an arbitrary set of values (e.g. $\{\text{AdS}, \text{dS}, \dots\}$). C is *normaliser-covariant* if $C(\pi) = C(\pi')$ whenever π and π' occur as $U_q(\mathfrak{su}(1,1))$ -constituents of one and the same irreducible unitary corepresentation of $\text{SU}_q(1,1) \rtimes \mathbb{Z}_2$ in the left regular corepresentation.

Theorem 1.21 (covariance obstruction). [PROVED] *Let C be any normaliser-covariant assignment. Then for every $\ell \in \frac{1}{2}\mathbb{Z}_{\geq 1}$, with $\epsilon = 0$ for integer ℓ and $\epsilon = \frac{1}{2}$ for half-odd ℓ ,*

$$C(\pi_{\ell-\frac{1}{2}, \epsilon}^S) = C(D_\ell^+) = C(D_\ell^-). \quad (10)$$

In particular no normaliser-covariant assignment separates the strange family at the Plancherel labels $a = \ell - \frac{1}{2} \in \frac{1}{2}\mathbb{Z}_{>0}$ from the discrete family: a criterion that classifies the discrete series as “AdS type” and the strange series as “dS type” cannot be normaliser-covariant.

Proof. Fix ℓ . For integer ℓ take $p = 1$ ($\chi(p) = 0$), $j = \ell$; for half-odd ℓ take $p = q$ ($\chi(p) = 1$), $j = \ell - \frac{1}{2}$. In both cases $\lambda = q^{1-2j}p^{-1} = q^{1-2\ell} > 1$, so Theorem 1.16 applies with $x = \mu(\lambda)$ and produces an irreducible corepresentation $W_{p,x}$ whose $U_q(\mathfrak{su}(1,1))$ -constituents are precisely $\pi_{\ell-1/2, \epsilon(p)}^S$, D_ℓ^- and D_ℓ^+ ; covariance forces (10). The parenthetical range statement: by Theorem 1.16(a) the strange labels occurring in the Plancherel decomposition are exactly $a = \ell - \frac{1}{2}$, $\ell \in \frac{1}{2}\mathbb{Z}_{\geq 1}$, i.e. $a \in \frac{1}{2}\mathbb{Z}_{>0}$ (Convention 2.6). $\square$

Corollary 1.22 (radial-problem version). [PROVED] *Let $\omega \in \mathbb{R}$ and consider the chord-lattice radial equation (R) of Section 4. The parity map $(c_k)_{k \geq 0} \mapsto ((-1)^k c_k)_{k \geq 0}$ is an involutive linear bijection from the solution space at ω onto the solution space at $-\omega$ which preserves absolute values pointwise — hence preserves square-summability of the normalised sequences, decay and growth rates, minimality of solutions, and maps the regular solution at ω (the one with $c_{-1} = 0$) to the regular solution at $-\omega$. Consequently any AdS/dS criterion that is a function of parity-invariant data of the radial problem assigns the same type to the spectral values $\omega = \mu(q^{2\ell-1})$ (discrete) and $\omega = -\mu(q^{2\ell-1})$ (strange).*

Proof. If $2\omega c_k = c_{k+1} + (1 - q^{2k})c_{k-1}$ for all $k \geq 1$, then $c'_k = (-1)^k c_k$ satisfies $2(-\omega)c'_k = (-1)^{k+1}2\omega c_k = (-1)^{k+1}c_{k+1} + (1 - q^{2k})(-1)^{k-1}c_{k-1} = c'_{k+1} + (1 - q^{2k})c'_{k-1}$. The map squares to the identity and $|c'_k| = |c_k|$; $c_{-1} = 0$ iff $c'_{-1} = 0$. All the listed properties depend on the sequences only through absolute values or through the vanishing of c_{-1} . $\square$

Remark 1.23 (reading: a candidate obstruction for Claim 1.8, and the consequence for issue #8). [ARGUED] Theorem 1.21 is an obstruction of exactly the kind admissible as outcome 3 of PROBLEM.md §6, *conditional on a covariance requirement*: if the criterion sought in issue #8 is required to be intrinsic to the quantum group structure that [1] argue is forced (“the reduction to the DSSYK model works exclusively at the level of the normaliser”), then it is normaliser-covariant, and by Theorem 1.21 it *cannot* classify the strange family as dS against an AdS discrete family. Any criterion validating Claim 1.8 must therefore break the $\mathbb{Z}_2$ -grading, i.e. use data beyond the normaliser-invariant structure. The only grading-breaking datum in the DSSYK setting identified so far is the orientation of the spectral axis — which sign of $\tilde{\Omega}$ is “low energy” — and that is precisely what the external dS proposal of Definition 1.3(a) supplies; a criterion using it distinguishes the families trivially (by the sign of $\tilde{\Omega}$), but its content then reduces to the hypothesis of Claim 1.7. Three further findings of this project point the same way: (1) the chord sector contains no strange *states* (Proposition 3.6), so no intrinsic spectral datum of the DSSYK sector sees the strange family directly; (2) the connection formula degenerates exactly at the Plancherel labels $a \in \frac{1}{2}\mathbb{Z}_{>0}$ (Theorem 4.2(c)), the same resonance structure as for the discrete family, not a new one; (3) no continuous parameter crosses anything (Remark 1.11). Issue #8 is accordingly reformulated (see its comment trail; the body edit is

recorded there verbatim and is pending repository write permissions, issue #17): its primary task is now to *construct the obstruction or refute it* — either exhibit a physically justified, grading-breaking criterion whose justification does not merely restate the dS identification, or establish that every reasonable criterion is covariant, which by Theorem 1.21 refutes Claim 1.8 as an intrinsic-geometry statement.

1.5 What the paper proves and what it argues

From the quotation ledger (`lit/notes/2512.10101.md` §3, §6): the operator-algebraic framework and the DSSYK reduction are derivations resting on published rigorous mathematics (Koelink–Kustermans, GKK); the strange-series kernels (4.19)–(4.20) are short derivations (re-derived above with two typos noted); the dS identification is an external *proposal*; the interpolation language is the authors’ gloss, explicitly conditional and explicitly left uninterpreted (“It is currently unknown what the correct physics interpretation of such an amplitude would be”); the classical-normaliser matrix elements are uncomputed (their §2.3 footnote), and the semigroup-Plancherel status is open (their §2.3 footnote). The project’s target is Claim 1.8; Claim 1.7 is its settled base point.

2 Conventions

This section fixes all conventions once; it agrees with `CONVENTIONS.md` exactly, and all sections below cite it. We follow the reference paper [1] (arXiv:2512.10101v2, henceforth “SI”) verbatim unless stated; “GKK” is Groenevelt–Koelink–Kustermans [6] (arXiv:0905.2830v2), “BINN” is [8], “BK” is Burban–Klimyk [9], “MMNNSU” is Masuda et al. [10]. Equation numbers of SI refer to the compiled v2. The numerical verification of every convention statement marked `[DERIVED]` below is in `src/check_conventions.py` (63/63 checks pass; output committed as `results/check_conventions.out`).

Convention 2.1 (Deformation parameter; DSSYK dictionary). q is real, $0 < q < 1$. The DSSYK chord parameter is $q_{\text{SYK}} := q^2$ [SI §4.2: “under the redefinition $q^2 \rightarrow q$ ”]; $q = e^{-\lambda/2}$, $q_{\text{SYK}} = e^{-\lambda}$ with $\lambda = 2p^2/N$ the DSSYK coupling [SI §4.1]. The energy of the principal-series eigenstate $\theta \in [0, \pi]$ is $E(\theta) = 2 \cos \theta / \sqrt{1 - q^2}$ [SI §4.2]. The lower spectral edge is $\theta = \pi$, the upper $\theta = 0$; the Schwarzschild/AdS₂ regime is $\theta = \pi - \ln q^2 |k|$, $q \rightarrow 1$ [SI (4.17)]; the de Sitter identifications appearing in the claim under test are recorded in Section 1.

Convention 2.2 ($U_q(\mathfrak{su}(1,1))$). Generators $E, F, K^{\pm 1}$ with [SI (3.3)–(3.4)]

$$KK^{-1} = K^{-1}K = 1, \quad KE = qEK, \quad KF = q^{-1}FK, \quad EF - FE = \frac{K^2 - K^{-2}}{q - q^{-1}}, \quad (11)$$

-structure $K^ = K$, $E^* = -F$; Hopf structure $\Delta(K) = K \otimes K$, $\Delta(E) = K \otimes E + E \otimes K^{-1}$, $\Delta(F) = K \otimes F + F \otimes K^{-1}$, $S(K) = K^{-1}$, $S(E) = -q^{-1}E$, $S(F) = -qF$, $\epsilon(K) = 1$, $\epsilon(E) = \epsilon(F) = 0$. Classically $K = q^{H/2}$.

Convention 2.3 (Casimir and transfer matrix).

$$\Omega := \frac{q^{-1}K^{-2} + qK^2 - 2}{(q - q^{-1})^2} + FE, \quad \tilde{\Omega} := \frac{1}{2}((q - q^{-1})^2\Omega + 2) = \frac{1}{2}(qK^2 + q^{-1}K^{-2}) + \frac{1}{2}(q - q^{-1})^2 FE \quad (12)$$

[SI (3.5) and §3.3, §3.5]. The DSSYK transfer matrix is $T = 2\tilde{\Omega}/\sqrt{1 - q^2}$ [SI §4.2], acting on chord states by $\tilde{\Omega}|k\rangle = \frac{1}{2}\sqrt{1 - q^{2k+2}}|k+1\rangle + \frac{1}{2}\sqrt{1 - q^{2k}}|k-1\rangle$ [SI (4.9)], with eigenvectors $|P^\theta\rangle = \sum_{k \geq 0} H_k(\cos \theta; q^2)/\sqrt{(q^2; q^2)_k}|k\rangle$ of eigenvalue $\cos \theta$ [SI (4.10)]. `[DERIVED]` (check C5). We write $\mu(y) := \frac{1}{2}(y + y^{-1})$ and $[z]_q := (q^z - q^{-z})/(q - q^{-1})$ [SI (A.1)].

Convention 2.4 (Coordinate algebra and the normaliser extension). $\mathcal{A}_q(\text{SU}(1,1))$: generators α, γ with relations, coproduct, counit, antipode as in SI (3.1)–(3.2). By Woronowicz’s no-go theorem there is no locally compact quantum group $\text{SU}_q(1,1)$; the extension $\text{SU}_q(1,1) \rtimes \mathbb{Z}_2$ adds a central group-like self-adjoint e , $e^2 = 1$, with $\alpha\alpha^* - q^2\gamma^*\gamma = e = \alpha^*\alpha - \gamma^*\gamma$ and the e -twisted coproduct [SI (3.14)–(3.15)]; its irreducible $*$ -representations π_θ act on $L_{q^2}^2(I_{q^2})$, $I_q = (-\infty, -1)_q \cup [0, \infty)_q$, with position operator $e\gamma^*\gamma$ [SI (3.18)]. The $\mathbb{Z}_2$ is the deck transformation between the two components $\text{SU}(1,1)$ and $\text{SU}(1,1)v$ of the classical normaliser of $\text{SU}(1,1)$ in $\text{SL}(2, \mathbb{C})$ [SI (2.25)]; the extension is needed because only for the normaliser do the self-adjoint extensions of $\Delta(e\gamma^*\gamma)$ admit coproduct-stable domains (“smoothness conditions”, SI (3.19)), making a von Neumann algebraic quantum group possible.

Convention 2.5 (Representation series; the strange series). Admissible $*$ -representations have K -eigenvalues in $q^{\frac{1}{2}\mathbb{Z}}$ with finite multiplicities [GKK §4.4]. The irreducible admissible unitary $*$ -representations, in our conventions (dictionary of Convention 2.8):

series	labels	K -spectrum	$\tilde{\Omega}$ eigenvalue
principal $\pi_{b,\epsilon}$	$b \in [0, -\frac{\pi}{2\ln q}]$, $\epsilon \in \{0, \frac{1}{2}\}$, $(b, \epsilon) \neq (0, \frac{1}{2})$	$q^{\mathbb{Z}+\epsilon}$	$\cos \theta$, $\theta = \pi + 2b \ln q$
discrete $D_\ell^\pm$	$\ell \in \frac{1}{2}\mathbb{Z}_{>0}$	$q^{\pm(\ell+\mathbb{Z}_{\geq 0})}$	$+\mu(q^{2\ell-1}) \in [1, \infty)$
complementary	$\lambda \in (-\frac{1}{2}, 0)$, $\epsilon = 0$	$q^{\mathbb{Z}}$	$+\mu(q^{1+2\lambda}) \in (1, \mu(q))$
strange $\pi_{a,\epsilon}^S$	$a > 0$, $\epsilon \in \{0, \frac{1}{2}\}$	$q^{\mathbb{Z}+\epsilon}$	$-\mu(q^{2a}) \in (-\infty, -1)$

Strange-series matrix elements on $\ell^2(\mathbb{Z})$ [GKK, eq. (strangeserrep)]:

$$K e_n = q^{n+\epsilon} e_n, \quad (q^{-1} - q) E_{\text{GKK}} e_n = q^{-n-\epsilon-\frac{1}{2}} \sqrt{(1+q^{2n+2\epsilon+1+2a})(1+q^{2n+2\epsilon+1-2a})} e_{n+1}, \quad (13)$$

unitary for *all* $a > 0$. SI generate the strange series from the discrete label by $\ell \rightarrow \ell + \frac{i\pi}{2\ln q}$ [SI §4.3]; the label map is $a = \ell - \frac{1}{2}$ (so $\ell = \frac{1}{2}$, i.e. $a = 0$, is the degenerate principal edge point, not a strange representation). *Decidability*: among irreducible admissible unitary $*$ -representations, $\tilde{\Omega} < -1$ holds exactly on the strange series. [DERIVED] (checks C1–C2; GKK §4.4 for completeness of the list).

Convention 2.6 (Regular representation and Plancherel decomposition). By GKK Thms. (decompKp $\pm\mp/\pm\pm$), the GNS space of the Haar weight of $L_q^\infty(\text{SU}(1,1) \rtimes \mathbb{Z}_2)$ decomposes as a $U_q(\mathfrak{su}(1,1))$ -module, per sector $(p \in q^{\mathbb{Z}}, \varepsilon, \eta)$, into the full principal integral $\int_0^{-\pi/2 \ln q} \pi_{b,\epsilon(p)} db$ plus discrete sums of discrete-series and of *strange-series* representations, the latter at $q^{2a} = pq^{1+2l}$ (hence $a \in \frac{1}{2}\mathbb{Z}_{>0}$); the complementary series is absent. $\epsilon(p) = \frac{1}{2}\chi(p) \bmod 1$, $\chi(p) = \log_q p$. [PROVED] (in GKK; not re-derived here).

Convention 2.7 (Chord sector; homogeneous-space objects). Chord states $|k\rangle_{\eta,\xi} = \lim_{r \rightarrow \infty} \delta_{q^r \eta, q^{2r} y, q^r \xi}^{-(st)}$, $y = q\eta\xi q^{2k}$, $k \in \mathbb{Z}_{\geq 0}$ [SI (4.5)]; the chord subsector sits at $\epsilon = -1$, the non-identity component of the normaliser. Rescaled spherical element $\hat{\rho}_{st}|k\rangle = \frac{1}{2}q^{-1}q^{-2k}|k\rangle$ [SI (4.13)]. Bilocal operators: discrete $\mathcal{O}^\ell(\tau) = e^{\tau T}(2q\hat{\rho}_{st})^{-\ell}e^{-\tau T}$ [SI (4.14)]; strange $\mathcal{S}^\ell(\tau) = e^{\tau T}(2q\hat{\rho}_{st})^{-\ell-\frac{i\pi}{2\ln q}}e^{-\tau T}$, $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$ [SI (4.18)]. Discrete-series overlap [SI (4.15)]:

$$\langle P^{\theta_1} | \mathcal{O}^\ell(0) | P^{\theta_2} \rangle = \frac{(q^{4\ell}; q^2)_\infty}{(q^{2\ell} e^{i(\pm\theta_1 \pm \theta_2)}; q^2)_\infty}, \quad (14)$$

and the strange-series overlap is (14) with $q^{2\ell} \rightarrow -q^{2\ell}$ [SI §4.3]. We write ℓ for the series label and ℓ_{length} for the bulk geodesic length (deviation from SI, who use ℓ for both; stated here once). The identification of the quantum homogeneous space itself is the subject of Section 3.

Convention 2.8 (Translation table). $E_{\text{SI}} = E_{\text{GKK}}$, $F_{\text{SI}} = -F_{\text{GKK}}$, $K_{\text{SI}} = K_{\text{GKK}}$, and $\tilde{\Omega} = -\Omega_{\text{GKK}}$ with $\Omega_{\text{GKK}} = \frac{1}{2}((q^{-1}-q)^2 F_{\text{GKK}} E_{\text{GKK}} - qK^2 - q^{-1}K^{-2})$ [DERIVED] (check C2). BINN [8]: $A = K$, $B = E$, $C = F$, $D = K^{-1}$, $\hat{q}_{\text{BINN}} = q$, $\Omega_{\text{BINN}} = \Omega$ [DERIVED] (check C3). MMNNSU: $\Omega_{\text{GKK}} = -\frac{1}{2}(q - q^{-1})^2 C_{\text{MMNNSU}} - 1$ [GKK, after their eq. (Casimir)], hence $C_{\text{MMNNSU}} = \Omega$. Askey–Wilson parameters of the Casimir action in the quantum Gauss decomposition: $(a, b, c, d) = (qs\eta, qt^{-1}\xi, qt\xi, qs^{-1}\eta)$ [SI §4.1]; the DSSYK sector is the $a, b, c, d \rightarrow 0$ (continuous q^2 -Hermite) limit. *Known printing discrepancy*: GKK state $\pi_{b,\epsilon}(\Omega_{\text{GKK}}) = \mu(q^{2ib})$; their printed matrix elements give $-\mu(q^{2ib}) = -\cos(2b \ln q)$. Resolved in favour of the matrix elements [DERIVED] (check C1, 60-digit arithmetic; the discrete- and strange-series values as printed *are* reproduced). The complementary-series value (unstated in GKK) is $\Omega_{\text{GKK}} = -\mu(q^{1+2\lambda})$, same check.

3 The quantum homogeneous space

3.1 Candidates

The phrase “quantum AdS_2 ” is used for inequivalent quotients in the literature; the computation of Sections 4–6 depends on which one is meant. The candidates appearing in the sources:

Definition 3.1 (quantum disk). The Cartan-type quotient $\text{SU}_q(1,1)/\text{U}(1)$ (invariant algebra generated by the spherical element $e\gamma^*\gamma$ in the normaliser setting; classically the unit disk, i.e. Euclidean AdS_2). This is the quantum bounded symmetric domain of Vaksman et al. Its transfer-matrix eigenfunctions are Al-Salam–Chihara-type (two levels up the q -Askey scheme from DSSYK). [1] §5 states explicitly that it is “another (distinct from the DSSYK coset considered in this paper) quantum homogeneous space of the von Neumann algebraic quantum group, corresponding to its quotient by a maximal compact subgroup.” *Not the DSSYK space.*

Definition 3.2 (quantum hyperboloid). The split-torus quotient (classically the single-sheeted hyperboloid = Lorentzian AdS_2). It does not occur in [1]’s reduction; we record it only to exclude it: no chord structure arises along this quotient in any source we know of. *Not the DSSYK space.*

Definition 3.3 (twisted-primitive double quotient; the DSSYK space). For s, t with $|s|, |t| > q$, the “ (s, t) -Gauss” double quotient of $\text{SU}_q(1,1) \rtimes \mathbb{Z}_2$ by the one-parameter quantum subgroups generated by the twisted primitive elements Y_s, Y_t [SI (3.43)], with invariant $*$ -algebra

$$\mathcal{A}_q((Y_s) \backslash \text{SU}(1,1) \rtimes \mathbb{Z}_2 / (Y_t)) = \langle \rho_{st}, e \rangle, \quad \rho_{st} := e\gamma_{s,qt}^* \gamma_{s,qt} - s^{-2} - q^{-2}t^{-2} \quad (15)$$

[SI (3.47)–(3.48)]. Functions on the total space diagonalising the symmetrised twisted primitives $\tilde{Y}_t^L, \tilde{Y}_s^R$ [SI (3.45)] with eigenvalues $\mu(\xi), \mu(\eta)$ (quantised momenta) and ρ_{st} -eigenvalue $\mu(y)$ are the mixed matrix elements $\delta_{\eta,y,\xi}^{\epsilon,(st)}$ [SI §3.4]; classically $\rho_{-1,1} = -2 + 4e^{2\varphi}$, so the ρ_{st} -eigenvalue is the radial/length coordinate [SI below (3.49)].

3.2 The choice, justified from the sources

Proposition 3.4 (where DSSYK lives). [DERIVED] (from [1] §4, whose derivations we follow; the chord-sector spectral statement is proved below.) *The DSSYK dynamics lives on the quotient of Definition 3.3, in the rescaled regime SI (4.2), as follows.*

- (a) Two-sided (chord) sector: *the span of the chord states $|k\rangle_{\eta,\xi}$, $y = q\eta\xi q^{2k}$, $k \in \mathbb{Z}_{\geq 0}$ [SI (4.1), (4.5)], which sit at $\epsilon = -1$ — the $\text{SU}(1,1)v$ component of the normaliser. On it the symmetrised Casimir acts as the DSSYK transfer matrix (Convention 2.3); DSSYK boundary conditions are $\eta = \xi = 1$.*

- (b) One-sided space (“reduced quantum $\text{AdS}_{2,q}$ ”): *fixing only the right datum $\xi \in q^{2\mathbb{Z}}$ gives the restricted q -lattice*

$$\mathbb{R}_{q^2}^2(\xi) = \{(-q^{2n}, q^{2m+1})\}_{n,m \in \mathbb{Z}} \cup \{(q^{2n}, q^{2m+1}) : q^{2(n+m)} \leq \xi\}, \quad (16)$$

with points (χ, y) , $\chi = -\epsilon\eta^{-1}$ [SI (4.22)], inner product

$$\langle f, g \rangle_\xi = \int_{-\infty}^{\infty} d_{q^2} \chi \int_0^{\infty} d_{q^2} y (q\xi^{-1} \chi y; q^2)_\infty \overline{f(\chi, y)} g(\chi, y) \quad (17)$$

[SI (4.24)], and Casimir action

$$\Omega = \frac{q^{-1} \mathcal{T}_q^{-2} + q \mathcal{T}_q^2 - 2}{(q - q^{-1})^2} - \mu_\xi y \frac{\chi}{1 - q^2} \mathcal{T}_q^{-2}, \quad \mu_\xi := \frac{\xi^{-1}}{1 - q^2}, \quad (18)$$

where $(\mathcal{T}_q f)(\chi, y) = f(\chi, qy)$ [SI (4.25)]. Classical coordinates: $p = \frac{1}{2}\chi/(q^{-1} - q)$, $e^\varphi = \frac{1}{2}y^{-1/2}$ [SI (4.30)]; the weight in (17) is the q -exponential $E_{q^2}(-2py\xi^{-1}(1-q^2)) \rightarrow e^{-2py\xi^{-1}}$ under asymptotic ($p_R \sim \lambda^{-1}$) boundary conditions, and $\rightarrow 1$ at fixed p_R , recovering classical AdS_2 data as $q \rightarrow 1$ [SI §4.5].

The DSSYK length dictionary: chord number k with $y = q\eta\xi q^{2k}$; renormalised length φ with $e^\varphi = \frac{1}{2}y^{-1/2}$; bare geodesic length $\ell_{\text{length}} = \lambda k$ in the triple-scaling regime, $\lambda = -2\ln q$; length positivity = the lattice restriction $q^{2(n+m)} \leq \xi$ [SI §4.1].

Remark 3.5 (position space is assumed, not constructed). [1] §4.4 constructs the space in momentum coordinates (χ, y) and writes: “we assume there exists an invertible q -Fourier transform” and “an appropriate explicit expression for the inner product in the position basis, which is compatible with the constraint and the inner product in the momentum space, is currently not available to us.” All computations in this project stay on the momentum-space lattice (16), where every object is defined; nothing below uses the assumed q -Fourier transform. If a later issue needs the position picture, that requires a gap issue first.

3.3 Where the strange series lives

Proposition 3.6 (chord-sector states are principal only). [PROVED] *The closure of the span of the chord states $\{|k\rangle\}_{k \geq 0}$ decomposes under the symmetrised Casimir $\tilde{\Omega}$ with purely absolutely continuous spectrum $[-1, 1]$ and spectral multiplicity one; i.e. only principal-series Casimir values occur, and no discrete, strange, or complementary values.*

Proof. $\tilde{\Omega}$ acts on $\ell^2(\mathbb{Z}_{\geq 0})$ as the Jacobi matrix with off-diagonal $a_k = \frac{1}{2}\sqrt{1 - q^{2k}}$ and zero diagonal (Convention 2.3). It is bounded and self-adjoint (all $|a_k| \leq \frac{1}{2}$), so its spectral measure at the cyclic vector $|0\rangle$ is the unique orthogonality measure of the corresponding orthonormal polynomials, which are the normalised continuous q^2 -Hermite polynomials; their orthogonality measure is $\frac{(q^2; q^2)_\infty}{2\pi} |(e^{2i\theta}; q^2)_\infty|^2 d\theta$ on $\cos \theta \in [-1, 1]$ — absolutely continuous with no mass points (Askey–Wilson $a = b = c = d = 0$; [7] §14.26). Cyclicity of $|0\rangle$ (three-term recurrence with $a_k > 0$ for $k \geq 1$) gives multiplicity one. By Convention 2.5, values in $[-1, 1]$ are principal-series values. $\square$

Proposition 3.7 (where the strange series does occur). [PROVED] (in [6]; not re-derived here.) *In the decomposition of the GNS space of the Haar weight of $L_q^\infty(\text{SU}(1, 1) \rtimes \mathbb{Z}_2)$ as a $U_q(\mathfrak{su}(1, 1))$ -module, the strange series occurs as discrete direct summands in the sectors $(\varepsilon, \eta) \in \{(+, -), (-, +), (+, +)\}$ with labels $q^{2a} = pq^{1+2l}$ (Convention 2.6).*

Remark 3.8 (consequence for the programme; the precise sense of “on the space”). Propositions 3.6 and 3.7 together fix what “matrix coefficients of the strange series on the DSSYK space” can mean, and this constrains issues #6–#8:

- (i) *As states of the DSSYK sector:* the strange series does **not** occur. The chord sector is principal only. Any interpolation mechanism through strange *states* of the transfer matrix is excluded.
- (ii) *As operators (bilocals):* the strange series acts on the chord sector through $\mathcal{S}^\ell$ (Convention 2.7), with the kernel properties of Section 1.
- (iii) *As functions on the group:* strange-series matrix coefficients are elements of $L_q^2(\mathrm{SU}(1,1) \rtimes \mathbb{Z}_2)$ (Proposition 3.7); their (s,t) -reduction obeys the radial Casimir equation (18) at spectral value $\tilde{\Omega} = -\mu(q^{2a}) < -1$, i.e. *below the principal band*. The radial problem is the same q -difference operator for every series; series membership enters only through the spectral value. This is the object issues #6–#7 compute.

4 Matrix coefficients of the strange series

Numerical verification for this section: `src/check_strange_coefficients.py` (output `results/check_strange` all checks pass; checks are cited as (S1)–(S5) below).

4.1 The representations, explicitly

Proposition 4.1 (strange series in our conventions). *[PROVED] (transcription of [6] eq. (strange-serrep) through the verified dictionary of Convention 2.8; checks C1–C2.) For $a > 0$, $\epsilon \in \{0, \frac{1}{2}\}$, the strange series $\pi_{a,\epsilon}^S$ of $U_q(\mathfrak{su}(1,1))$ acts on $\ell^2(\mathbb{Z})$ by*

$$\begin{aligned} K e_n &= q^{n+\epsilon} e_n, \\ (q^{-1} - q) E e_n &= q^{-n-\epsilon-\frac{1}{2}} \sqrt{(1+q^{2n+2\epsilon+1+2a})(1+q^{2n+2\epsilon+1-2a})} e_{n+1}, \\ F &= -E^*, \end{aligned} \tag{19}$$

with $\tilde{\Omega} = -\mu(q^{2a})$, equivalently $\Omega = -2(\mu(q^{2a}) + 1)/(q - q^{-1})^2 = -(q^a + q^{-a})^2/(q - q^{-1})^2$ (Convention 2.3). It is unitary (all square-root arguments are manifestly positive for every $a > 0$), irreducible, and admissible; the normaliser extension acts by the $\mathbb{Z}_2$ -grading operators U_0^{+-}, U_0^{-+} of [6] which map $\tilde{\Omega} \mapsto -\tilde{\Omega}$, i.e. exchange the strange family with the discrete family and the principal band with itself [SI §3.3]; the precise statements, including the fusion of $\pi_{\ell-1/2}^S$ with $D_\ell^\pm$ into a single irreducible corepresentation, are Proposition 1.15 and Theorem 1.16.

4.2 Radial coefficients on the chord lattice

By Remark 3.8, the meaningful “matrix coefficients on the DSSYK space” are the solutions of the radial Casimir equation at the strange spectral value. On the chord lattice the radial equation is the three-term relation of Convention 2.3; in unnormalised form ($c_k = v_k \sqrt{(q^2; q^2)_k}$):

$$2\omega c_k = c_{k+1} + (1 - q^{2k}) c_{k-1}, \quad k \geq 1, \tag{R}$$

with the *regular* solution fixed by $c_1 = 2\omega c_0$ (equivalently $c_{-1} = 0$), i.e. $c_k^{\mathrm{reg}} = H_k(\omega; q^2)$.

Theorem 4.2 (strange radial coefficients, closed forms). *Let $0 < s < 1$ and $\omega = -\mu(s) < -1$ (strange values: $s = q^{2a}$, $a > 0$; Convention 2.5).*

(a) *[PROVED] (minimal solution.) The sequence*

$$c_k^S(a) := (-1)^k q^{2ak} {}_2\varphi_1 \left(\begin{matrix} 0, 0 \\ q^{4a+2} \end{matrix}; q^2, q^{2k+2} \right), \quad {}_2\varphi_1 \left(\begin{matrix} 0, 0 \\ c \end{matrix}; q, z \right) = \sum_{j \geq 0} \frac{z^j}{(q; q)_j (c; q)_j}, \tag{20}$$

solves (R) at $\omega = -\mu(q^{2a})$ for all $k \geq 1$, and $c_k^S(a) = (-1)^k q^{2ak} (1 + O(q^{2k}))$; the normalised sequence $v_k^S = c_k^S / \sqrt{(q^2; q^2)_k}$ is square-summable. (Check S1'; proof below.)

- (b) *[DERIVED]* (regular solution and growth.) The regular solution is $c_k^{\text{reg}} = H_k(-\mu(q^{2a}); q^2) = (-1)^k H_k(\mu(q^{2a}); q^2)$ with $H_k(\mu(s); q^2) s^k (s^2; q^2)_\infty \rightarrow 1$ as $k \rightarrow \infty$; it grows like q^{-2ak} and is not square-summable — consistently with Proposition 3.6 (no strange states). (Check S2.)
- (c) *[DERIVED]* (Wronskian and connection formula.) With $W_k := c_{k+1}^{\text{reg}} c_k^{\text{min}} - c_{k+1}^{\text{min}} c_k^{\text{reg}}$ one has $W_k = (q^2; q^2)_k W_0$ (*[PROVED]*, one-line induction from (R)), and

$$W_0 = \frac{s^{-1} - s}{(s^2; q^2)_\infty (q^2; q^2)_\infty} \neq 0 \quad \text{for all } 0 < s < 1, \quad (21)$$

equivalently the connection formula (generic s , $s^2 \notin q^{2\mathbb{Z}}$)

$$H_k(\mu(s); q^2) = \frac{c_k^{\text{min}}(s^{-1})}{(s^2; q^2)_\infty} + \frac{c_k^{\text{min}}(s)}{(s^{-2}; q^2)_\infty}, \quad c_k^{\text{min}}(s) := s^k {}_2\varphi_1 \left(\begin{matrix} 0, 0 \\ s^2 q^2 \end{matrix}; q^2, q^{2k+2} \right), \quad (22)$$

verified to relative 10^{-33} or better at multiple (q, s) (checks S3, S3', S5). At the resonant points $s^2 \in q^{2\mathbb{Z}}$ — which include every strange label occurring in the Plancherel decomposition, $a \in \frac{1}{2}\mathbb{Z}_{>0}$ — the branch $c^{\text{min}}(s^{-1})$ degenerates and (22) requires a confluent limit; the minimal solution (20) itself remains well-defined there.

Proof of (a). Substitute $c_k = s^k \varphi(q^{2k})$, $\varphi(z) = \sum_{j \geq 0} d_j z^j$, into (R) at $\omega = \mu(s)$ (the value $-\mu(s)$ follows by the parity map $c_k \rightarrow (-1)^k c_k$, which flips the sign of ω in (R)). Matching coefficients of z^j in $(s^2 + 1)\varphi(z) = s^2 \varphi(q^2 z) + (1 - z)\varphi(q^{-2} z)$ gives $d_j [(s^2 + 1) - s^2 q^{2j} - q^{-2j}] = -d_{j-1} q^{2-2j}$, i.e. $d_j = d_{j-1} q^2 / [(1 - q^{2j})(1 - s^2 q^{2j})]$, so $d_j = q^{2j} / [(q^2; q^2)_j (s^2 q^2; q^2)_j]$ with $d_0 = 1$ — the series (20). For $0 < s < 1$ the factors $(1 - s^2 q^{2i})$ are bounded below by $1 - s^2 q^2 > 0$, so $|d_j| \leq C q^{2j}$ and φ is analytic in $|z| < q^{-2}$, which contains every lattice point $z = q^{2k}$, $k \geq 0$; the term-by-term manipulation is therefore justified, and $\varphi(q^{2k}) = 1 + O(q^{2k})$ gives the stated asymptotics and square-summability. (Numerical confirmation to 10^{-40} : checks S1, S1'). $\square$

Remark 4.3 (the ρ -shift). At the discrete-series value $\omega = \mu(q^{2\ell-1})$ the minimal solution decays like $q^{(2\ell-1)k}$ while the discrete bilocal (Convention 2.7) weights chord states by $q^{2\ell k}$; the mismatch q^{-k} , constant = 1 numerically (check S4), is the usual Weyl-vector shift between an operator weight and a wavefunction decay rate, recorded here so that it is not mistaken for an inconsistency downstream. The same shift appears in the strange family: bilocal weight $(-1)^k q^{2\ell k}$ versus minimal-solution decay $(-1)^k q^{2ak} = (-1)^k q^{(2\ell-1)k}$ under $a = \ell - \frac{1}{2}$.

4.3 Bilocal coefficients between principal states

The equal-time overlap kernels of the strange bilocals are Lemma 1.4: for $\ell \in \frac{1}{2}\mathbb{Z}_{>0}$ the chord sum converges absolutely ($|t| = q^{2\ell} < 1$) and

$$\langle P^{\theta_1} | \mathcal{S}^\ell(0) | P^{\theta_2} \rangle = \frac{(q^{4\ell}; q^2)_\infty}{(-q^{2\ell} e^{i(\pm\theta_1 \pm \theta_2)}; q^2)_\infty}, \quad (23)$$

holomorphic in $q^{2\ell}$ on the unit disk; no analytic continuation from any compact-quotient computation is used anywhere in this section (everything is defined directly on the non-compact lattice), so the caveat of arXiv:2412.19681 about continuation-based derivations does not apply here.

4.4 Orthogonality and Plancherel: what is available and what is not

Proposition 4.4 (available). *[PROVED]* (in the sources.) (i) On the chord sector, the principal eigenfunctions are complete and delta-orthonormal with the DSSYK Plancherel density

$\frac{d\theta}{2\pi}(q^2, e^{\pm 2i\theta}; q^2)_\infty$ [SI (4.11); Proposition 3.6]. (ii) At the unrescaled level, the GKK sector decompositions (Convention 2.6) provide a rigorous Plancherel theory in which the strange series occurs with counting measure at $q^{2a} = pq^{1+2l}$.

Remark 4.5 (not available: rescaled Plancherel for the strange family). [GAP: ISSUE #14] No orthogonality or completeness statement for strange-series coefficients on the DSSYK space is available: the strange radial coefficients (20) are solutions at spectral values off the spectrum of the self-adjoint transfer matrix, and whether the GKK Plancherel decomposition survives SI's rescaling $r \rightarrow \infty$ is not established — by us or by [1]. Every statement in this project that would sum or integrate over strange labels on the DSSYK space is conditional on issue #14 and says so. The unconditional statements are: the GKK theory at finite scaling, the chord-sector principal completeness, the kernel computations of Section 1, and Theorem 4.2.

5 The radial part of the Casimir

The method fixed by `PROBLEM.md` §4 is the Casimir-radial-part technique of Schlösser–Isachenkov [11] (classical, via Matsuki decomposition), whose use for conformal blocks (Casimir $\rightarrow$ radial part $\rightarrow$ Calogero–Sutherland/Pöschl–Teller) is developed in [12, 13]; this section transports it to the quantum side, starting from the momentum-space Casimir action (18) (Remark 3.5: everything stays on the momentum lattice). Numerical verification for this section: `src/check_radial_casimir.py` (output `results/check_radial_casimir.out`, 45/45 checks pass; cited as (R0)–(R7) below).

5.1 Slice reduction: the radial operator

Proposition 5.1 (slice reduction of the momentum-space Casimir). [PROVED] (computation; checks (R1).) Let $q \in (0, 1)$, $\xi \in q^{2\mathbb{Z}}$, and let Ω act on functions on the restricted lattice $\mathbb{R}_{q^2}^2(\xi)$ of (16) by (18), with the inner product (17). Then:

- (a) (slices) Ω commutes with multiplication by χ , hence preserves each slice $\{(\chi, q^{2m+1})\}_m$; with $f_m := f(\chi, q^{2m+1})$ the symmetrised Casimir $\tilde{\Omega} = \frac{1}{2}((q - q^{-1})^2 \Omega + 2)$ acts on the slice by the three-term q -difference operator

$$(\tilde{\Omega}f)_m = \frac{q}{2} f_{m+1} + \frac{1}{2q} (1 - cq^{2m}) f_{m-1}, \quad c := \chi/\xi \in \pm q^{2\mathbb{Z}}, \quad (24)$$

which depends on the slice only through c .

- (b) (symmetry) (24) is symmetric with respect to the slice weight induced by (17), $w_m = q^{2m+1}(cq^{2m+2}; q^2)_\infty$: one has $w_m \cdot \frac{q}{2} = w_{m+1} \cdot \frac{1}{2q} (1 - cq^{2m+2})$ for all m . For $c = q^{2\nu} > 0$ the weight vanishes identically for $m < m_0 := -\nu$ (an exactly vanishing factor of the q -Pochhammer): the length-positivity truncation of Proposition 3.4 is carried by the measure.

- (c) (orthonormal form) In the orthonormalised basis $g_m = \sqrt{w_m} f_m$, $\tilde{\Omega}$ acts as the Jacobi operator

$$(J_c g)_m = a_m g_{m+1} + a_{m-1} g_{m-1}, \quad a_m = \frac{1}{2} \sqrt{1 - cq^{2m+2}}, \quad (25)$$

with zero diagonal, on $\ell^2(\mathbb{Z})$ for $c < 0$ and on $\ell^2(\{m \geq m_0\})$ for $c > 0$.

- (d) (two radial operators) The unit shift $m \mapsto m + 1$ intertwines J_c with J_{cq^2} . Hence, up to unitary equivalence, exactly two radial operators occur on the whole space: for every slice with $\chi\xi > 0$ (the $\epsilon = -1$, chord/normaliser wedge), $J_+ := J_{c>0}$ is the DSSYK transfer matrix $\tilde{\Omega}$ of Convention 2.3 under $k = m - m_0$; and for every slice with $\chi\xi < 0$ (the $\epsilon = +1$ wedge), $J_- := J_{c=-1}$, with $a_m = \frac{1}{2} \sqrt{1 + q^{2m+2}}$, $m \in \mathbb{Z}$.

Proof. (a) (18) contains only y -shifts $\mathcal{T}_q^{\pm 2}$ and multiplication operators; $\frac{1}{2}(q-q^{-1})^2$ times the potential term gives, at $y = q^{2m+1}$, the coefficient $\frac{1}{2}(q-q^{-1})^2 \xi^{-1} \chi y / (1-q^2)^2 = \frac{1}{2} q^{-2} \chi \xi^{-1} q^{2m+1} = \frac{1}{2q} c q^{2m}$, since $(q-q^{-1})^2 / (1-q^2)^2 = q^{-2}$; the kinetic part contributes $\frac{q}{2} f_{m+1} + \frac{1}{2q} f_{m-1}$ after the $+1$ from the symmetrisation cancels the $-f_m$. (b) $w_m \frac{q}{2} = \frac{1}{2} q^{2m+2} (c q^{2m+2}; q^2)_\infty$ and $w_{m+1} \frac{1}{2q} (1 - c q^{2m+2}) = \frac{1}{2} q^{2m+2} (1 - c q^{2m+2}) (c q^{2m+4}; q^2)_\infty$, equal by $(u; q^2)_\infty = (1-u)(u q^2; q^2)_\infty$; for $c = q^{2\nu}$ and $m \leq -\nu - 1$ the argument $c q^{2m+2} \in q^{-2\mathbb{Z}_{\geq 0}}$ makes a factor of $(c q^{2m+2}; q^2)_\infty$ vanish exactly. (c) $a_m = \frac{q}{2} \sqrt{w_m / w_{m+1}} = \frac{1}{2} \sqrt{1 - c q^{2m+2}}$. (d) Substituting $m = k + m_0$, $c = q^{2\nu}$, $m_0 = -\nu$ into (25) gives $a = \frac{1}{2} \sqrt{1 - q^{2k+2}}$, the chord coefficients; all statements are verified numerically to 10^{-60} or exactly in checks (R1). $\square$

Remark 5.2 (the effective potential; where the representation label sits). The radial “potential” is the single q -exponential coefficient

$$1 - c q^{2m} = 1 - 4 p p_R (q^{-1} - q)^2 y / q, \quad y = q^{2m+1}, \quad e^\varphi = \frac{1}{2} y^{-1/2}, \quad (26)$$

using SI’s dictionary $p = \frac{1}{2} \chi / (q^{-1} - q)$, $p_R = \frac{1}{2} \xi^{-1} / (q^{-1} - q)$ [SI (4.30), (4.4)]: a q -Liouville (one-exponential) potential, the complete $a, b, c, d \rightarrow 0$ degeneration of the Askey–Wilson (q -Pöschl–Teller) operator SI (3.85) whose parameters are $(q s \eta, q t^{-1} \xi, q t \xi, q s^{-1} \eta)$ (Convention 2.8). Issue #7’s task “write the effective potential as a function of the strange-series label” is therefore answered in the negative direction the data force: *the potential carries no representation label*. The label enters the radial problem only (i) as the spectral value $\omega = \tilde{\Omega}$ -eigenvalue ($-\mu(q^{2a})$ strange, $+\mu(q^{2\ell-1})$ discrete, $\cos \theta$ principal) — as already recorded in Remark 3.8(iii) — and (ii) as the boundary decay exponent of the radial solution, $g_m^{(\omega)} \sim q^{1/2} \sigma^m / (\sigma^2 q^2; q^2)_\infty$ as $m \rightarrow +\infty$ with $\sigma = -q^{2a}$, resp. $q^{2\ell-1}$ (Lemma 5.6).

5.2 The chord wedge

Proposition 5.3 (the $\chi \xi > 0$ wedge). [PROVED] *On every slice with $\chi \xi > 0$ the radial operator is the chord transfer matrix $\tilde{\Omega}$: bounded, self-adjoint, with purely absolutely continuous spectrum $[-1, 1]$ of multiplicity one and no eigenvalues (Proposition 3.6). In particular no strange or discrete Casimir values occur on the chord wedge, for any slice and any boundary datum ξ .*

Proof. Proposition 5.1(d) and Proposition 3.6. (Check (R5), control part: Dirichlet and Neumann truncations of the chord wedge both have zero out-of-band eigenvalues.) $\square$

5.3 The classical limit

Theorem 5.4 ($q \rightarrow 1$ contraction of the radial operator). [DERIVED] (checks (R2).) *Fix $p, p_R \in \mathbb{R} \setminus \{0\}$ and set $\chi = 2p(q^{-1} - q)$, $\xi^{-1} = 2p_R(q^{-1} - q)$, $\lambda = -2 \ln q$, and $e^\varphi = \frac{1}{2} y^{-1/2}$ [SI (4.4), (4.30)]. Then, pointwise on smooth functions of φ and with deviation $O(\lambda^2)$,*

$$\frac{2(J_c - 1)}{\lambda^2} \longrightarrow h_{\text{cl}} := \frac{1}{4} \partial_\varphi^2 - p p_R e^{-2\varphi}, \quad (27)$$

the Liouville radial operator: a wall for $p p_R > 0$ (the chord wedge) and an inverted (bottomless) exponential well for $p p_R < 0$ (the $\epsilon = +1$ wedge). Equivalently, in terms of SI’s classical radial Casimir action (2.26) [label `eq:classicalCasimirAction`], $(-\Omega_{\text{cl}})(\varphi) = (-\frac{1}{4} \partial_\varphi^2 - \frac{1}{2} \partial_\varphi + p_L p_R e^{-2\varphi}) f$ on $L^2(e^{2\varphi} d\varphi)$: conjugation by e^φ gives the unitary equivalence $-\Omega_{\text{cl}} \cong \frac{1}{4} - h_{\text{cl}}$, so (27) reproduces SI’s classical radial part with the band-edge constant $\frac{1}{4}$ accounted for (principal eigenvalues: $-\Omega_{\text{cl}} = \frac{1}{4} + s^2 \leftrightarrow h_{\text{cl}}\text{-value} - s^2$). Moreover:

- (a) (both edges) *The contraction (27) captures the upper spectral edge ($\tilde{\Omega} \approx +1$); the lower (Schwarzian/AdS) edge is the same classical operator reached through the grading unitary: exactly, at every q , $-(\tilde{\Omega} + 1) \circ \Pi = \Pi \circ (\tilde{\Omega} - 1)$ on each slice (Lemma 1.13; check (R2), deviation 0), so $-2(\tilde{\Omega} + 1)/\lambda^2$ on the $(-1)^m$ -twisted sector contracts to the same h_{cl} .*

- (b) (eigenfunction check) *On the wall side, h_{cl} has the generalised eigenfunctions $K_{2is}(2\sqrt{pp_R}e^{-\varphi})$ with h_{cl} -eigenvalue $-s^2$ (the Liouville/Schwarzian wavefunctions of SI (2.27)); numerically, $2(J_c - 1)/\lambda^2$ applied to the sampled Bessel function returns the eigenvalue with deviation 3.9×10^{-3} , 8.0×10^{-4} , 1.8×10^{-4} at $\lambda = 0.2, 0.1, 0.05$ (ratios 4.9, 4.4; predicted rate 4), and the operator-level deviations on test functions have measured ratios 4.8, 4.3 ($pp_R > 0$) and 4.3, 4.2 ($pp_R < 0$) per halving of λ (checks (R2)).*
- (c) (gauge caveat, recorded) *In the unnormalised function picture (24) the same contraction produces the similarity-equivalent form $\frac{1}{4}(1 - \partial_\varphi)^2 - pp_R e^{-2\varphi}$ but with an $O(\lambda)$ midpoint artifact in pointwise comparisons; the gauge-invariant $O(\lambda^2)$ statement is (27) in the orthonormal gauge. (First run of check (R2) was written in the function gauge and FAILED with rate ratios ≈ 2 ; recorded per CONTRIBUTING.md §5.)*

Derivation. With $\ln y$ -derivative $D = y\partial_y = -\frac{1}{2}\partial_\varphi$ one has $g(q^{\pm 2}y) = e^{\mp \lambda D}g$, so the kinetic part of (25) is $\frac{1}{2}(e^{\lambda D} + e^{-\lambda D})g - g = \frac{\lambda^2}{2}D^2g + O(\lambda^4)$, giving $2(\cdot)/\lambda^2 \rightarrow D^2 = \frac{1}{4}\partial_\varphi^2$. The potential part: $a_m - \frac{1}{2} = -\frac{1}{4}cq^{2m+2} + O(c^2q^{4m})$ with $cq^{2m+2} = 4pp_R(q^{-1} - q)^2qy = 4pp_R\lambda^2y(1 + O(\lambda^2))$, and the two neighbours contribute qy and $q^{-1}y$ whose average is $y \cosh(\lambda/2) \cdot (1 + O(\lambda^2))$, so $2(a_m(g_{m+1}) + a_{m-1}(g_{m-1}) - \text{kinetic})/\lambda^2 \rightarrow -4pp_Ry = -pp_R e^{-2\varphi}$ with $O(\lambda^2)$ error. The unitary equivalence with SI (2.26): for $u = e^\varphi f$, $e^\varphi(-\frac{1}{4}\partial^2 - \frac{1}{2}\partial + V)e^{-\varphi}u = (-\frac{1}{4}\partial^2 + \frac{1}{4} + V)u$. (a) is Lemma 1.13(ii) restricted to a slice (J_c has zero diagonal, so $(-1)^m$ -conjugation flips its sign); (b) is the standard Bessel computation, $\frac{1}{4}\partial_\varphi^2 K_{2is}(ue^{-\varphi}) = (u^2 e^{-2\varphi}/4 - s^2)K_{2is}$, verified numerically as stated. $\square$

Remark 5.5 (comparison with the Matsuki-decomposition machinery). [ARGUED] Issue #7 asks whether the classical Matsuki picture of [11] has a q -analogue here. What the computation delivers is the q -analogue of *its output*, not of the decomposition theory itself: classically, the double reduction by the two parabolic (twisted-primitive limit) subgroups splits the group into the two open regions $p_L p_R \geq 0$, on each of which the Casimir has the radial part SI (2.26) — the Liouville wall/well pair; on the quantum side, Proposition 5.1 splits the momentum lattice into the two slice families $\chi\xi \geq 0$ ($\epsilon = \mp 1$), carrying J_+ (wall, truncating) and J_- (well, non-truncating), and Theorem 5.4 shows each contracts onto its classical counterpart. The classical radial coordinate ranges over a continuum touching the orbit boundaries; the q -lattice $\pm q^{2\mathbb{Z}} \times q^{2\mathbb{Z}+1}$ never meets the walls $\chi = 0, y = 0$. We do *not* construct a q -Matsuki decomposition theory (no claim about general double cosets is made or needed: SI's quantum Gauss decomposition already supplies the reduction); this is the stated deviation-with-justification from PROBLEM.md §4's method sketch. SI themselves note the agreement of (18) with the hands-on lattice Casimir of BINN [8] (4.10) [SI §4.4].

5.4 The non-chord wedge: solutions, deficiency indices, extensions

Throughout this subsection $c = -\gamma < 0$ with $\gamma \in q^{2\mathbb{Z}}$; by Proposition 5.1(d) we may and do take $\gamma = 1$, i.e. J_- with $a_m = \frac{1}{2}\sqrt{1 + q^{2m+2}}$ on $\ell^2(\mathbb{Z})$; general γ statements follow by the shift. The unnormalised picture is

$$2\omega c_m = c_{m+1} + (1 + \gamma q^{2m}) c_{m-1}, \quad m \in \mathbb{Z}, \quad (\text{R}')$$

the analytic continuation of the chord recurrence (R) under $q^{2k} \mapsto -\gamma q^{2m}$, i.e. under the *same* half-period shift $k \mapsto m + \frac{\ln \gamma}{2 \ln q} + \frac{i\pi}{2 \ln q}$ that generates the strange series (Convention 2.5).

Lemma 5.6 (closed-form solutions on the wedge). [DERIVED] (checks (R0), (R3).) For $\sigma \in \mathbb{C}^\times$, $|\sigma| < 1$, and $\omega = \mu(\sigma)$:

(a) the sequence $c_m(\sigma) := \sigma^m {}_2\varphi_1\left(\begin{smallmatrix} 0, 0 \\ \sigma^2 q^2 \end{smallmatrix}; q^2, -\gamma q^{2m+2}\right)$ solves (R') wherever the series converges ($\gamma q^{2m+2} < 1$); the derivation of Theorem 4.2(a) carries over verbatim with $q^{2k} \rightarrow -\gamma q^{2m}$;

(b) by the $a \rightarrow 0, b \rightarrow 0$ confluence of Heine's transformation [[14] (1.4.1)],

$${}_2\varphi_1\left(\begin{smallmatrix} 0, 0 \\ C \end{smallmatrix}; q, Z\right) = \frac{{}_1\varphi_1\left(\begin{smallmatrix} Z \\ 0 \end{smallmatrix}; q, C\right)}{(C; q)_\infty (Z; q)_\infty}, \quad {}_1\varphi_1\left(\begin{smallmatrix} Z \\ 0 \end{smallmatrix}; q, C\right) = \sum_{n \geq 0} \frac{(Z; q)_n}{(q; q)_n} (-1)^n q^{\binom{n}{2}} C^n, \quad (28)$$

whose right side is entire in Z ; hence

$$c_m(\sigma) = \frac{\sigma^m {}_1\varphi_1\left(\begin{smallmatrix} -\gamma q^{2m+2} \\ 0 \end{smallmatrix}; q^2, \sigma^2 q^2\right)}{(\sigma^2 q^2; q^2)_\infty (-\gamma q^{2m+2}; q^2)_\infty} \quad (29)$$

extends $c_m(\sigma)$ to every $m \in \mathbb{Z}$ (the denominator never vanishes for $\gamma > 0$), and $c_m(\sigma) = \sigma^m(1 + O(q^{2m}))$ as $m \rightarrow +\infty$;

(c) the weighted solution is $g_m^{(\omega)} = q^{1/2} \sqrt{(-\gamma q^{2m+2}; q^2)_\infty} c_m(\sigma)$.

The identity (28) and the solution property of (29) at $m \leq 0$ are verified to 10^{-57} and 10^{-158} relative (checks (R0), (R3)).

Theorem 5.7 (deficiency indices of the wedge radial operator). [PROVED] (in [15]; parameter identification ours; numerically verified, checks (R4), (R5).) The operator J_- , defined on finitely supported sequences in $\ell^2(\mathbb{Z})$, is unbounded, symmetric, and not essentially self-adjoint: its deficiency indices are $(1, 1)$. Specifically, $2J_-$ is the operator L of [15], case (1) of his Lemma 3.1 with $\psi = 0$: his $a_k = |r|^{-1} q^{-k} \sqrt{1 - 2rq^k \sin \psi + r^2 q^{2k}}$ satisfies $a_k^2 = 1 + r^{-2} q^{-2k}$ at $\psi = 0$, which equals our $(2a_m)^2 = 1 + \gamma q^{2m+2}$ under $e_m \mapsto e_{-m}$ and $r = \gamma^{-1/2}$ (the bond $(m, m+1)$ maps to his bond $(-m-1, -m)$). The structure of the two ends:

(a) (horizon end, limit circle) As $m \rightarrow -\infty$ (i.e. $y \rightarrow \infty, \varphi \rightarrow -\infty$) every solution of (R') is square-summable in the weighted norm, with $|g_m| \sim \text{const} \cdot q^{|m|/2}$; measured slopes of $\ln_q |g_m|$ over $m \in [-32, -20]$: 0.500008 and 0.500008 for the two basis solutions at a strange-type ω , 0.499989, 0.499989 at $\omega = i$ (predicted $\frac{1}{2}$, tolerance 1.8×10^{-3} from the $O(q^{|m|})$ correction scale; checks (R4)).

(b) (boundary end, limit point) As $m \rightarrow +\infty$ the coefficients are bounded ($a_m \rightarrow \frac{1}{2}$), so the half-line restriction is bounded and self-adjoint there; for $\omega \notin [-1, 1]$ exactly one solution ($c_m(\sigma) \sim \sigma^m, |\sigma| < 1$) decays, the other grows like σ^{-m} .

Consequently the eigenvalue equation at $\omega = i$ has exactly one $\ell^2(\mathbb{Z})$ solution: deficiency indices $(1, 1)$, and the self-adjoint extensions $J_-^{(\theta)}$ form a $U(1)$ family.

Remark 5.8 (first analysis, and why it was wrong). Our first asymptotic analysis of the horizon end concluded limit point (a Gaussian/ anti-Gaussian solution pair), through a mis-estimated reduction-of-order sum; the correct statement is Koelink's: in the unnormalised picture all solutions decay at the same *Gaussian* leading order $q^{m^2+O(m)}$ (the recurrence degenerates to a one-term recursion downward), the weight grows like $q^{-(m+O(1))^2}$, and the products sit at the common *exponential* order $q^{|m|/2}$ — both square-summable. Recorded per CONTRIBUTING.md §5; the numerical slopes above are the check that settled it. The extension dependence is also directly visible numerically: out-of-band eigenvalues of Dirichlet vs. Neumann truncations at $m = -M$ remain separated by gaps in $[0.11, 0.84]$ as $M = 8, 12, 16$ grows (check (R5)) — for an essentially self-adjoint operator they would converge to each other.

Theorem 5.9 (spectra of the self-adjoint extensions). *Let $J_-^{(\theta)}$, $\theta \in \mathbb{U}(1)$, be the self-adjoint extensions of J_- .*

- (a) [PROVED] (in [15], Thm. 5.8 and Rmk. 5.11 (arXiv labels: `thm:case1spectraldecomp`, `rmk:discspectralE`), transcribed through the dictionary of Theorem 5.7.) *Every $J_-^{(\theta)}$ has absolutely continuous spectrum $[-1, 1]$ of multiplicity one, plus simple point spectrum*

$$\sigma_{\text{pp}}(J_-^{(\theta)}) = \mu\left((y_1(\theta)q^{2\mathbb{Z}} \cup y_2(\theta)q^{2\mathbb{Z}}) \cap \{|y| > 1\}\right), \quad (30)$$

two bilateral q^2 -grids, unbounded in both directions, determined by the two roots y_1, y_2 (per fundamental cell) of the equation $g(y) = \text{const}(\theta)$ for an explicit elliptic function g of order two (a quotient of two theta functions; proof of (b) below); the eigenfunction at $\omega = \mu(y_i q^{2j})$ is the explicit $g^{(\omega)}$ of Lemma 5.6(c).

- (b) [DERIVED] (root product rule.) *The two roots obey*

$$y_1 y_2 \in -\gamma q^{2\mathbb{Z}}. \quad (31)$$

Consequently y_1 and y_2 are real of opposite signs: every self-adjoint extension carries one infinite discrete-type tower ($\tilde{\Omega} = \mu(y) > 1$) and one infinite strange-type tower ($\tilde{\Omega} = \mu(y) < -1$), log-periodically locked by (31). In particular, on the $\gamma = 1$ slice the extension whose spectrum contains the strange value $-\mu(q^{2a_0})$ ($a_0 > 0$) has

$$\sigma_{\text{pp}} = \{-\mu(q^{2a_0+2j})\}_{j \in \mathbb{Z}, q^{2a_0+2j} < 1} \cup \{+\mu(q^{2j-2a_0})\}_{j \in \mathbb{Z}, q^{2j-2a_0} < 1} \quad (32)$$

(the two parametrisations $\mu(y) = \mu(1/y)$ are used with arguments inside $(0, 1)$, so the $|y| > 1$ window of (30) is automatic).

- (c) [DERIVED] (numerical verification.) *With the bilinear eigenvector test $h(\omega) = \sum_m \hat{g}_m^{(\omega_0)} \hat{g}_m^{(\omega)}$ (two real normalisable solutions belong to one extension iff $h = 0$; Green's identity), at $q = 0.6$, $\gamma = 1$: for $a_0 = \frac{1}{2}$, $|h|$ at the four predicted grid points $-\mu(q^3), -\mu(q^5), +\mu(q), +\mu(q^3)$ is $7.8 \times 10^{-42}, 1.3 \times 10^{-41}, 1.4 \times 10^{-41}, 7.8 \times 10^{-42}$ (the numerical floor of the m -cutoffs), against off-grid controls 0.209 and 0.097; for generic $a_0 = 0.37$, the strange grid point and the two product-rule partners give $|h| \leq 1.3 \times 10^{-41}$ while the naive mirror points $+\mu(q^{2a_0}), +\mu(q^{2a_0+2})$ give 0.043, 0.063: the partner grid is (32), not the mirror. (Checks (R6); grid persistence at $q = 0.9$, check (R7).)*

Proof of (b). In Koelink's variables (case (1), $\psi = 0$, $a = q^{1/2}$, $t = ir$, $r = \gamma^{-1/2}$, base q with our q^2 -grids arising as his " q^2 -quadratic grids") the point-spectrum condition is $A\theta(yq^{1/2}ir) + \bar{A}\theta(-yq^{1/2}ir) = 0$ with $\theta(x) = (x, q/x; q)_\infty$ and $A = A(\theta)$ [[15], Rmk. 5.11, display eq:weightE1]. Writing $y = e^{2\pi iw}$, $q = e^{\pi i\tau}$, the function $g(w, \tau) = \theta(e^{2\pi iw}q^{1/2}ir)/\theta(-e^{2\pi iw}q^{1/2}ir)$ is elliptic of order 2 with periods $1, \tau$ (i.e. $y \mapsto yq^2$), and the condition is the level set $g(w, \tau) = -\bar{A}/A$. By Abel's theorem the two roots w_1, w_2 of a level set of an elliptic function satisfy $w_1 + w_2 \equiv$ (sum of the poles) mod the period lattice; the poles of g are at $-yq^{1/2}ir \in q^{\mathbb{Z}}$, two per cell, with product $y_{p1}y_{p2} = (-q^{-1/2}/ir)(-q^{1/2}/ir) = -1/r^2 = -\gamma$; hence $y_1y_2 \equiv -\gamma \bmod q^{2\mathbb{Z}}$. A complex-conjugate root pair would have $y_1y_2 = |y_1|^2 > 0$, contradicting (31); so both roots are real, of opposite signs. (32) is (31) solved for y_2 at $y_1 = -q^{2a_0}$. $\square$

Corollary 5.10 (parity on the wedge; the fusion extension). [DERIVED] (checks (R6d).) $\Pi J_- \Pi = -J_-$ exactly (J_- has zero diagonal), so Π maps the extension family to itself with $\sigma(J_-^{(\theta')}) = -\sigma(J_-^{(\theta)})$, pairing the towers as in Corollary 1.22. The extension through $-\mu(q^{2a_0})$ is Π -invariant iff its two grids are mirrors of each other, i.e. iff $2a_0 \in \mathbb{Z}$ — exactly the Plancherel labels $a_0 \in \frac{1}{2}\mathbb{Z}_{>0}$ of Convention 2.6. At those labels its spectrum is

$$\sigma_{\text{pp}} = \{\pm\mu(q^{2a_0+2j})\}_{j \in \mathbb{Z}, q^{2a_0+2j} < 1}, \quad \text{i.e. strange labels } a \in (a_0 + \mathbb{Z})_{>0} \text{ and discrete labels } \ell = a + \frac{1}{2} : \quad (33)$$

the radial problem realises the fusion pairing $a = \ell - \frac{1}{2}$ of Theorem 1.16 spectrally, discrete and strange towers inside one self-adjoint operator. For $2a_0 \notin \mathbb{Z}$ the mirror points are not in the spectrum (measured $|h| = 0.043, 0.063$ vs. 10^{-41} on-grid; check (R6)).

Remark 5.11 (which extension is physical: open). [GAP: ISSUE #26] Nothing computed here selects one θ . At the unrescaled level GKK’s Casimir is *canonically* self-adjoint — the closure is unique by strong commutation of the closed generators, and where a sector Jacobi operator fails to be essentially self-adjoint (their $(+, +)$ sector at $m = 0$) the extension is fixed by affiliation to the dual von Neumann algebra $\hat{M}$ [GKK, proof of Lemma lem:commutationEE*pmepet and Thm. thm:Jacobioperatorforsec=q: the domain condition $u_{-k} = q^{k/2}f(q^k)$, f differentiable at 0]. SI’s rescaling (4.2) acts on states and on $\hat{\rho}_{st}$, not on $\hat{M}$ or on operator domains; whether it carries GKK’s canonical domain to a distinguished θ — conjecturally the Π -invariant fusion extension (33) — is issue #26 (the domain-level counterpart of issue #16). Until it is resolved, every statement about “the” point spectrum of the wedge radial Casimir is per-extension.

Remark 5.12 (horizon reading; agreement with BINN). BINN [8] met the same operator family from the hands-on lattice side and drew the physical boundary condition we cannot yet derive: for their non-truncating transfer matrices (same recurrence (R’)-type, studied via [15]), “the resulting operator has a continuous one-parameter family of self-adjoint extensions”, “the spectrum always contains some bound states”, and — verbatim — “We interpret the bound states that appear on top of that as corresponding to operators sitting at the Poincaré horizon”, with the truncating choice justified by demanding “that no additional states are localized just on the Poincaré horizon” [BINN §4.4]. In the von Neumann algebraic construction the truncating wedge is not a *choice* but one of the two slice families ($\chi\xi > 0$), and the non-truncating wedge ($\chi\xi < 0$, the $\epsilon = +1$ component) is genuinely present, with the horizon bound states (30) at discrete- and strange-type Casimir values. Two differences from BINN’s misaligned case are recorded: their non-truncating operator has $\chi > 0$ off the lattice alignment (coefficient $1 - \chi q^{2m}$ changing sign), ours has $\chi < 0$ (coefficient $1 + \gamma q^{2m} > 0$ throughout); and in GKK’s unrescaled sector operators the corresponding point spectra are *asymmetric* (infinitely many strange-type, finitely many discrete-type eigenvalues in the $(+, +)$ sector, GKK §8.3) — the diagonal term that causes the asymmetry dies under SI’s rescaling, which restores the exact Π -symmetry of Corollary 5.10.

5.5 The strange series in the radial problem, and the $q \rightarrow 1$ limit

Theorem 5.13 (where the strange values sit radially). [DERIVED] (from Theorems 5.7–5.9; checks (R6).) *On the DSSYK space, the strange Casimir values $\omega = -\mu(q^{2a}) < -1$ occur in the radial problem exactly as follows.*

- (i) *On the chord wedge ($\chi\xi > 0$): never (Proposition 5.3).*
- (ii) *On the $\epsilon = +1$ wedge ($\chi\xi < 0$): for every $a > 0$ the explicit radial solution $g^{(-\mu(q^{2a}))}$ of Lemma 5.6 is normalisable (geometric decay $|\sigma|^m = q^{2am}$ at the boundary end, $q^{|m|/2}$ at the horizon end), and $-\mu(q^{2a})$ is a simple eigenvalue of exactly one self-adjoint extension $J_-^{(\theta(a))}$; within that extension it comes locked to the discrete-type tower (32). The discrete values $+\mu(q^{2\ell-1})$ enter in the mirror-symmetric way. The strange and discrete families are therefore exactly on a par in the radial problem; no radial datum distinguishes them except the sign of $\tilde{\Omega}$ — the concrete realisation of Corollary 1.22 and Theorem 1.21 announced there.*

Proposition 5.14 ($q \rightarrow 1$: how the strange towers disappear). [DERIVED] (checks (R7).) *Fix $a > 0$ and let $q \rightarrow 1^-$.*

- (i) The strange eigenvalue $-\mu(q^{2a}) \rightarrow -1^-$: its distance to the continuum $[-1, 1]$ is $\mu(q^{2a}) - 1 = 2 \sinh^2(a\lambda/2) = \frac{a^2\lambda^2}{2} + O(\lambda^4)$ — the strange towers merge into the band edge and no isolated spectrum survives at $q = 1$ at fixed label. The strange representation disappears (its Casimir parameter leaves the strange range $(-\infty, -1)$), as required by *PROBLEM.md*.
- (ii) In lower-edge (Schwarzian) units, $E^{\text{low}} := 2(1 + \tilde{\Omega})/\lambda^2$ on the Π -twisted sector (Theorem 5.4(a)): the continuum becomes $[0, \infty)$ ($E^{\text{low}} \rightarrow k^2$) and the strange tower survives at

$$E^{\text{low}}(-\mu(q^{2a_0+2j})) = \frac{2(1 - \mu(q^{2a_0+2j}))}{\lambda^2} \longrightarrow -(a_0 + j)^2 : \quad (34)$$

the bound-state tower of the classical inverted-Liouville well. Its $-\Omega_{\text{cl}}$ values are $\frac{1}{4} - (a_0 + j)^2 = \ell_j(1 - \ell_j)$ with $\ell_j = a_0 + j + \frac{1}{2}$: the classical discrete-series Casimir values, under the fusion label map $a = \ell - \frac{1}{2}$ once more. Measured deviations from $-(a_0 + j)^2$ at $a_0 = \frac{1}{2}$: $(-2.3 \times 10^{-4}, -1.9 \times 10^{-2}, -1.5 \times 10^{-1})$ for $j = 0, 1, 2$ at $q = 0.9$ and $(-2.1 \times 10^{-6}, -1.7 \times 10^{-4}, -1.3 \times 10^{-3})$ at $q = 0.99$; deviation ratio 109.9 against the predicted $(\ln 0.9 / \ln 0.99)^2 = 109.9$.

- (iii) At the same (lower) edge the discrete tower is expelled: $E^{\text{low}}(+\mu(q^{2\ell-1})) = 2(1 + \mu(q^{2\ell-1}))/\lambda^2 = 4/\lambda^2 + (\ell - \frac{1}{2})^2 + O(\lambda^2) \rightarrow +\infty$ (measured: 90.33 vs $4/\lambda^2 = 90.08$ at $q = 0.9$; 9900.33 vs 9900.08 at $q = 0.99$). At the upper edge the roles are exactly exchanged (Π -conjugation). So each classical edge theory keeps one tower — which becomes its ordinary discrete series — and expels the mirror tower to infinite energy; the coexistence of both towers in one self-adjoint radial operator is a finite- q ($\lambda > 0$) phenomenon with no classical counterpart.
- (iv) **[ARGUED]** (a quantum-only window.) The classical well $\frac{1}{4}\partial_\varphi^2 - pp_R e^{-2\varphi}$, $pp_R < 0$, on $L^2(e^{2\varphi} d\varphi)$ admits bound states only for $\kappa > \frac{1}{2}$ (normalisability of $e^{(2-4\kappa)\varphi}$ at $\varphi \rightarrow +\infty$, $E = -\kappa^2$), matching the classical unitarity bound $\ell = \kappa + \frac{1}{2} > 1$; the q -side towers (34) include $\kappa = a_0 + j \in (0, \frac{1}{2})$ members whenever $a_0 < \frac{1}{2}$, whose limits fall below that bound: eigenvalues with no classical counterpart even edge-locally. Not needed downstream; recorded for issue #8.

Remark 5.15 (summary for the criterion (issue #8)). The radial part of the Casimir on the DSSYK space is now explicit: the chord wedge carries the transfer matrix (principal continuum only); the $\epsilon = +1$ wedge carries the q -Liouville well operator J_- , whose self-adjoint realisations each hold one strange and one discrete tower, locked by the product rule (31), collapsing at $q \rightarrow 1$ onto the classical discrete series of the respective edge with the mirror tower expelled. Everything is exactly Π -covariant; the only non-covariant datum remains the orientation of the spectral axis (which edge is “low”), and the only unresolved freedom is the extension choice (issue #26), whose conjectured resolution (33) is itself Π -invariant. Any AdS/dS criterion built from the radial problem therefore faces the obstruction of Theorem 1.21 in the sharpened form: it must break a symmetry that the radial operator, its extension family, its explicit eigenfunctions, and its classical limits all respect.

6 The AdS/dS criterion

This section executes issue #8 as reformulated under issue #15 (Remark 1.23): assess the candidate criteria, construct the obstruction or refute it, and decide the verdict for Claim 1.8. *Provenance, for the acceptance requirement that the criterion not be chosen to fit the answer:* the candidate list below is the one fixed in issue #8’s body at project start, and the covariance obstruction (Theorem 1.21) was proved under issue #15 — both *before* the radial computation of Section 5 existed.

6.1 Criterion classes and the candidate list

Definition 6.1 (criterion classes). An AdS/dS criterion is an assignment C from the irreducible admissible unitary $*$ -representations of $U_q(\mathfrak{su}(1,1))$ (equivalently, from their occurrences in the radial problem of Section 5) to values in $\{\text{AdS}, \text{dS}, \dots\}$, built from:

- (I) *intrinsic data*: the quantum group, the DSSYK space, the radial operators, their extension theory, eigenfunctions and classical limits — i.e. normaliser-covariant data in the sense of Definition 1.20;
- (II) *intrinsic data plus the orientation*: the identification of the lower spectral edge $\theta = \pi$ as the ground-state (Schwarzian/AdS) side — intrinsic to DSSYK as a thermal theory ($e^{-\beta T}$ weights low energy; Definition 1.2), and the one datum in the model that is odd under the grading Π ;
- (III) class (II) data *plus an external bulk proposal* — the edge-dS or centre-dS identifications of Definition 1.3.

The candidates fixed in issue #8: (a) the sign of the effective curvature extracted from the radial operator; (b) the sign of the exponential term in the (q -)Pöschl–Teller potential; (c) the character of the spectrum (continuous vs. discrete, normalisable vs. delta-normalisable); (d) reality/unitarity conditions on the representation label; (e) the signature of the invariant metric on the quantum homogeneous space.

6.2 Assessment of the candidates

Proposition 6.2 (candidates (a)–(d) cannot validate Claim 1.8). [PROVED] (from Section 5 and Theorem 1.21.) *Each of the criteria (a)–(d), evaluated on the DSSYK radial problem, is a normaliser-covariant assignment; by Theorem 1.21 it gives $C(\pi_{\ell-1/2}^S) = C(D_\ell^\pm)$ and therefore cannot classify the strange family as dS against an AdS discrete family. Case by case:*

- (a) Effective curvature. *The radial potential (26) carries no representation label (Remark 5.2); its classical limits on the two wedges are the AdS_2 Liouville wall and well of SI (2.26) (Theorem 5.4) — both wedges are regions of the same classical AdS_2 reduction. No curvature-like coefficient anywhere in the radial family depends on which representation is inserted: criterion (a) is constant on representations.*
- (b) Potential sign. *The sign of the exponential term is $\text{sgn}(\chi\xi)$ — a wedge datum (which slice), not a representation datum. Constant on representations.*
- (c) Spectral character. *Strange and discrete values enter identically: simple isolated eigenvalues of the same wedge extensions, in towers exchanged by Π with all normalisability data preserved (Theorem 5.9, Corollary 1.22); principal values are the common continuum. The one apparent intrinsic asymmetry — strange radial solutions alternate at lattice scale, $c_m \sim (-1)^m q^{2am}$, so they have no continuum limit as functions — is precisely the parity twist, i.e. it is equivalent to $\text{sgn} \tilde{\Omega}$, whose reading is the orientation question of class (II), not an intrinsic discriminator.*
- (d) Label reality/unitarity. *Both families are unitary at every label (Convention 2.5; the strange series for all $a > 0$), and the fusion theorem pairs them label-by-label inside single irreducible corepresentations (Theorem 1.16). Covariant, hence obstructed.*

Remark 6.3 (candidate (e): not constructible in the present framework). A metric-signature criterion cannot be evaluated: the DSSYK space is constructed in momentum coordinates, the position picture rests on an assumed q -Fourier transform (Remark 3.5), and a metric requires

the spectral-triple layer that [1] explicitly defer: “the natural metric structure provided by a spectral triple should shed light on the intriguing overlap of correlation functions between AdS and dS regions of the q -deformed homogeneous space, which we observed in Section 4.3. [...] Implementing and exploring this vision in the present context is a work in progress” [SI §5]. Nothing in this project assumes such a metric (so this is not a **gap** in our argument); but it is the one listed candidate that could in principle carry grading-breaking structure, and Conjecture 6.6 records what it would have to do.

6.3 The oriented data

Proposition 6.4 (what the orientation separates). *[DERIVED] (arithmetic from Section 5; checks (R7).) Relative to the DSSYK orientation (class (II)): on every wedge extension $J_-^{(\theta)}$, the strange tower lies below the lower band edge — at energies $E = 2\tilde{\Omega}/\sqrt{1-q^2}$ a distance*

$$\Delta E_a = \frac{2(\mu(q^{2a}) - 1)}{\sqrt{1-q^2}} = \frac{4\sinh^2(a\lambda/2)}{\sqrt{1-e^{-\lambda}}} \quad (35)$$

below the chord-sector ground-state energy $-E_{\max}$ — and the discrete tower symmetrically above $+E_{\max}$. As $q \rightarrow 1$ the strange tower survives at the AdS (ground-state) edge as the classical discrete-series bound states of the inverted-Liouville well, and is expelled from the opposite edge; the discrete tower has the mirror fate (Proposition 5.14). The orientation therefore does break the grading and does separate the two families — but it attaches the strange family to the AdS side and the discrete family to the opposite edge. (These towers live on the $\epsilon = +1$ wedge of the reduced space, not in the chord sector, whose spectrum remains exactly the band $[-E_{\max}, E_{\max}]$: Proposition 3.6; there is no contradiction with DSSYK spectral data.)

6.4 Verdict

Theorem 6.5 (verdict on Claim 1.8). *Under the criterion classes of Definition 6.1:*

- (i) *[PROVED] (class (I): refuted.) No intrinsic criterion validates Claim 1.8: the listed candidates (a)–(d) fail by Proposition 6.2, and any intrinsic criterion is normaliser-covariant, hence obstructed by Theorem 1.21; candidate (e) is not constructible (Remark 6.3).*
- (ii) *[DERIVED] (class (II): refuted with an inversion.) The orientation separates the families, but assigns the strange family to the AdS side (Proposition 6.4): under the edge-dS reading of the opposite edge, the dS-edge-local family is the discrete series. An oriented criterion therefore not only fails to validate “the strange series realises the dS₂ type” — it supports the opposite assignment.*
- (iii) *[DERIVED] (class (III): reduces to Claim 1.7.) Under edge-dS, what is true is exactly Proposition 1.6: the strange bilocals are the edge-crossing family (finite AdS $\leftrightarrow$ dS amplitudes, vanishing AdS $\leftrightarrow$ AdS ones, the reverse for discrete bilocals) — with the selectivity refined on the complex-weight family by the transition of Theorem 6.8 (Section 6.5), and still open only for the non-spherical bilocals (issue #13, Remark 1.12). Under centre-dS, the strange AdS $\leftrightarrow$ dS amplitude vanishes (Remark 1.10) and no spectral statement singles out the band centre: the claim fails in every class.*
- (iv) *[DERIVED] (No crossing.) In no class does any quantity cross as the strange label a varies: the eigenvalue moves monotonically along $(-\infty, -1)$, the bilocal amplitudes vary smoothly, and the only distinguished labels are the resonances $2a \in \mathbb{Z}$ (Plancherel labels, Π -invariant extensions, connection-formula degeneration (Theorem 4.2(c))) — structure conditions, not sign changes. The “interpolation” of Claim 1.8, read as a crossing against any continuous parameter (Remark 1.11), does not occur.*

Conclusion for the epic. *Claim 1.8 is **refuted** as an intrinsic-geometry statement, by the explicit obstruction now realised in the radial problem; the defensible content of “the strange series interpolates between AdS and dS” is Claim 1.7 (conditional on edge-dS), together with the structural statement that the strange and discrete families form a single Π -orbit — one fused irreducible corepresentation of the normaliser (Theorem 1.16), one locked pair of spectral towers per radial extension (Theorem 5.9), the two towers edge-local at the two ends of the spectrum with the strange component on the AdS side and the cross-edge amplitudes carried by the strange bilocals. The $\text{AdS} \leftrightarrow \text{dS}$ bridge, to the extent the framework supports one, is the grading Π ; the strange series is one half of its orbit, not a family that migrates from one geometry to the other.*

Claim 6.6 (the falsifiable residue). *[CONJECTURED] If the spectral-triple metric structure announced in [1] §5 is constructed for the reduced space, then: (1) it will be Π -covariant on the intrinsic data, and its AdS/dS decomposition will enter through the spectral orientation, i.e. through the same external input as Definition 1.3; and (2) evaluated on the wedge towers it will attach the strange tower to the AdS-edge region and the discrete tower to the opposite edge region, as in Proposition 6.4. Reasons: (1) any structure built functorially from the quantum group data inherits the covariance of Theorem 1.21; (2) is the $q \rightarrow 1$ spectral localisation of Proposition 5.14, which a compatible metric must reproduce. Either part is falsifiable by the future construction; failure of (1) would exhibit exactly the grading-breaking structure whose absence drives Theorem 6.5, and would reopen Claim 1.8.*

6.5 The complex-weight bilocal family: an exact edge-connectivity transition

The selectivity load on Theorem 6.5(iii) can be discharged for the largest family the framework makes computable: the *complex-weight* spherical-power bilocals.

Lemma 6.7 (complex-weight bilocals; exact reflection). *[PROVED] (checks (P1).) For $w \in \mathbb{C}$ with $\text{Re } w > 0$, the operator $(2q\hat{\rho}_{st})^{-w}$ is a bounded diagonal bilocal with chord weights t^k , $t := q^{2w}$, $0 < |t| < 1$, and its equal-time kernel between principal states is the q -Mehler kernel (2) at complex t (absolutely convergent; both sides of (2) are analytic in t on the unit disk). The reflection identity holds for all such t : $K_q(t; \theta_1, \theta_2) = K_q(-t; \theta_1, \pi - \theta_2)$, by the proof of Lemma 1.5 verbatim; hence the normalisation-free edge ratio*

$$\text{Rat}(t) := \frac{K_q(t; \theta_q^L(k_1), \theta_q^U(k_2))}{K_q(t; \theta_q^L(k_1), \theta_q^L(k_2))} = \prod_{j=1}^4 \frac{(te^{i\varphi_j}; q^2)_\infty}{(-te^{i\varphi_j}; q^2)_\infty}, \quad \varphi_j \in \{\pm\lambda_2(k_1+k_2), \pm\lambda_2(k_1-k_2)\}, \quad (36)$$

$\lambda_2 = |\ln q^2|$, satisfies $\text{Rat}(t)\text{Rat}(-t) = 1$ identically. On the principal-weight circle $t = q^{1+2ib}$ ($\Delta = \frac{1}{2} + ib$; $b \in [0, b_{\max}]$, $b_{\max} = -\pi/(2 \ln q)$, the parameter window of the principal series), the reflection $b \mapsto b_{\max} - b$ sends $t \mapsto -\bar{t}$, so $|\text{Rat}(b_{\max} - b)| = 1/|\text{Rat}(b)|$: the family is exactly antisymmetric about its midpoint.

Theorem 6.8 (the edge-connectivity transition). *Fix $k_1, k_2 \geq 0$ and $\delta \in (0, \frac{\pi}{2})$; let $t = \rho_0 e^{i\alpha}$ with $\rho_0 \in [\frac{1}{2} + \delta, 1)$, and let $q \rightarrow 1^-$.*

- (i) *[PROVED] (suppressed half-disk.) If $|\alpha| + \lambda_2(k_1+k_2) \leq \frac{\pi}{2} - \delta$, then $|\text{Rat}(t)| \rightarrow 0$: each factor of (36) obeys the counting bound of Proposition 1.6(iv) with $c_0 = \cos(|\alpha| + \lambda_2(k_1+k_2)) \geq \sin \delta$ in place of $\frac{1}{2}$ (the per-factor ratio $g_c(\rho) = (1 - 2\rho c + \rho^2)/(1 + 2\rho c + \rho^2)$ is decreasing in ρ for $c > 0$, derivative numerator $2c(\rho^2 - 1) < 0$, so $g_{c_0}(\rho_n) \leq g_{c_0}(\frac{1}{2}) < 1$ on the $\rightarrow \infty$ many sites with $\rho_n = \rho_0 q^{2n} \geq \frac{1}{2}$). Quantitatively, $\ln |\text{Rat}| = \sum_j \text{Re}[\text{Li}_2(-te^{i\varphi_j}) - \text{Li}_2(te^{i\varphi_j})]/\lambda_2 + \frac{1}{2} \sum_j \ln \left| \frac{1-te^{i\varphi_j}}{1+te^{i\varphi_j}} \right| + O(\lambda_2^2)$; at $|t| \rightarrow 1$ the leading exponent per factor is $-(\frac{\pi^2}{4} - \frac{\pi|\alpha|}{2})/\lambda_2$. Measured at $\alpha = \pi/4$, $\rho_0 = q$: $\ln |\text{Rat}| = -23.4186, -81.0067, -245.504$ against predictions $-23.3979, -81.0058, -245.504$ at $q = 0.9, 0.97, 0.99$; residual scaling exponent measured ≈ 2 in λ_2 (checks (P2)).*

- (ii) **[PROVED]** (enhanced half-disk.) *If $|\alpha| - \lambda_2(k_1 + k_2) \geq \frac{\pi}{2} + \delta$, then $|\text{Rat}(t)| \rightarrow \infty$, reciprocally to the mirror point: $\text{Rat}(t) = 1/\text{Rat}(-t)$ exactly. (Measured: $|\text{Rat}| = 1.5 \times 10^{10}$ at $q = 0.9$, 4.2×10^{106} at $q = 0.99$ for $\alpha = 3\pi/4$; checks (P3).) The strange ray $\alpha = \pi$ ($t = -q^{2\ell}$) is the boundary ray of maximal enhancement at given $|t|$.*
- (iii) **[PROVED]** (the marginal line is exact.) *For $t = i\rho$, $\rho \in (0, 1)$, at every q : $|\text{Rat}(i\rho)| = 1$ — the cross-edge and same-edge kernels have exactly equal modulus. Proof: the phase set $\{\varphi_j\}$ is symmetric, and for each pair $(\varphi, -\varphi)$ and each $n \geq 0$, $|1 \mp \rho_n e^{i(\pi/2 + \varphi)}|^2 |1 \mp \rho_n e^{i(\pi/2 - \varphi)}|^2 = (1 \pm 2\rho_n \sin \varphi + \rho_n^2)(1 \mp 2\rho_n \sin \varphi + \rho_n^2)$ is the same for both signs, so numerator and denominator products in (36) have equal modulus factor by factor. (Verified to 5×10^{-41} ; checks (P4).)*
- (iv) **[PROVED]** (dictionary: the crossing sits exactly at the band centre.) *On the principal-weight circle $t = q^{1+2ib}$ the point $b = b_{\max}/2$ has $t = -iq$, on the marginal line; hence $|\text{Rat}| = 1$ there exactly, with $\ln|\text{Rat}| < 0$ below and > 0 above (antisymmetric by Lemma 6.7; measured ∓ 9.367 , ∓ 32.4 , ∓ 98.2 at $0.4b_{\max}/0.6b_{\max}$ for $q = 0.9, 0.97, 0.99$; checks (P5)). The θ -image of the crossing weight ($\theta = \pi + 2b \ln q$ at $b = b_{\max}/2$) is $\theta = \pi/2$ — the band centre, at every q . (A first version of check (P5) searched for an asymptotic drift of the crossing towards the midpoint and found none: the measured drift equalled bisection precision at every q ; the crossing is exact, not asymptotic. Recorded per CONTRIBUTING.md §5.)*

Remark 6.9 (reading of the transition). **[ARGUED]** Three consequences. (1) *Binary selectivity fails for the extended family*: the edge-connecting set of the weight disk is the open half $|\arg t| > \pi/2$, with the strange ray as its extreme; Claim 1.7’s selectivity statement is exact for DSSYK’s standard matter content (real positive weights and their strange partners) and becomes the *transition* of Theorem 6.8 on the full family. Theorem 6.5(iii) is refined accordingly. (2) *The transition locus is the Π -fixed locus, and its θ -image is the band centre* — the locus of the centre-dS identification (Definition 1.3(b)). Under edge-dS, edge-connectivity switches on exactly at the weight whose spectral image is the rival proposal’s “dS point”. We record this structural coincidence linking the two dS proposals without interpreting it. (3) *What remains outside this family*: bilocals transforming in $\pi_{b,e}$ as genuine non-spherical corepresentation intertwiners — the operators SI describe below their (4.14) as spanned by $\Lambda^{-1}(\delta^{\epsilon,(st)})$ -elements, whose action they found “impractical” to compute — are not powers of the spherical element and are not covered; issue #13 remains open for them.

Remark 6.10 (what remains open after this section). The verdict is unconditional on the radial side but leaves three loads on other issues: the selectivity of C-amp beyond the spherical-power family (issue #13; partially resolved by Theorem 6.8, issue #27); the rescaled Plancherel theory (issue #14) for any statement summing over strange labels; and the extension selection (issue #26), which affects only *which* locked tower pair is realised, not the verdict (every extension is obstructed the same way, and the conjectured canonical extension is itself Π -invariant). The centre-dS quantitative comparison deferred from Remark 1.10 is subsumed by Theorem 6.5(iii) and Theorem 6.8.

7 Limits and verification

The cross-check targets fixed by PROBLEM.md §5 include, besides the q -Askey and dS-holography papers recorded in Section 1, the sine-dilaton picture [16] and the von Neumann algebra results of [17] (type II_1 factor); the corresponding comparisons are the business of issue #11.

7.1 The $q \rightarrow 1$ dictionary

The four acceptance limits of PROBLEM.md §5, with their status:

Proposition 7.1 (density of states). *[DERIVED] (checks (L1).) The chord-sector Plancherel density [SI (4.11); Proposition 4.4] factorises exactly at every q through the q -Gamma function: at $\theta = \pi - \lambda_2 \tilde{k}$,*

$$\rho_q(\theta) := \frac{(q^2; q^2)_\infty}{2\pi} |(e^{2i\theta}; q^2)_\infty|^2 = \frac{(q^2; q^2)_\infty^3 (1 - q^2)^2}{2\pi} \frac{1}{|\Gamma_{q^2}(2i\tilde{k})|^2} \xrightarrow{q \rightarrow 1} \frac{(\dots)}{2\pi} \cdot \frac{2\tilde{k} \sinh(2\pi\tilde{k})}{\pi}, \quad (37)$$

the Schwarzian density of states, by $\Gamma_{q^2} \rightarrow \Gamma$ and $|\Gamma(2i\tilde{k})|^2 = \pi/(2\tilde{k} \sinh 2\pi\tilde{k})$. The factorisation is verified to 10^{-37} relative; the limit with relative deviations 0.054/0.0056/0.00056 at $q = 0.9/0.99/0.999$ ($\tilde{k} = 0.6$) and 0.47/0.057/0.0058 ($\tilde{k} = 1.4$), measured rate ratios 0.78–0.99 against the predicted $O(\lambda_2)$ scaling.

Proposition 7.2 (wavefunctions: Bessel K from the connection formula). *[DERIVED] (checks (L2); derivation from this project's own closed forms.) Let $\tilde{s} > 0$, $s = q^{2i\tilde{s}}$ (principal value $\mu(s) = \cos(\tilde{s}\lambda_2)$), and $u = 2q^{k+1}/\lambda_2$ (the renormalised-length variable, $u = e^{-\tilde{\ell}/2}$ with $\tilde{\ell} = \lambda_2 k - 2\ln(\lambda_2/2) + O(\lambda_2)$, cf. the length dictionary of Proposition 3.4). Then, as $q \rightarrow 1$ at fixed $(\tilde{s}, u)$:*

(a) *the minimal-solution kernel of Theorem 4.2 conflues to the modified Bessel function,*

$${}_2\varphi_1\left(\begin{matrix} 0, 0 \\ s^2 q^2 \end{matrix}; q^2, \frac{\lambda_2^2 u^2}{4}\right) \longrightarrow \Gamma(1 + 2i\tilde{s}) (u/2)^{-2i\tilde{s}} I_{2i\tilde{s}}(u), \quad (38)$$

(measured relative deviations $6.6 \times 10^{-3}/5.8 \times 10^{-4}/5.7 \times 10^{-5}$ at $q = 0.9/0.99/0.999$, $u = 0.4$; rates 1.10, 1.01 vs. predicted $O(\lambda_2)$);

(b) *combining the two branches through the connection formula (22) and $\Gamma(1 + \nu)\Gamma(-\nu) = -\pi/\sin \pi\nu$, $K_\nu = \frac{\pi}{2\sin \pi\nu}(I_{-\nu} - I_\nu)$:*

$$H_k(\mu(q^{2i\tilde{s}}); q^2) \longrightarrow \frac{2 K_{2i\tilde{s}}(u)}{\lambda_2 (q^2; q^2)_\infty}, \quad (39)$$

i.e. the transfer-matrix eigenvectors reduce to the Liouville/Schwarzian wavefunctions $K_{2is}(e^{-\tilde{\ell}/2})$ of SI (2.27) at the upper edge, and times $(-1)^k$ (Lemma 1.13) at the lower/AdS edge. Measured relative deviations 0.16/0.014/0.0014 ($u_0 = 0.4$, oscillatory region, k up to 7819) and 0.089/0.0081/0.0008 ($u_0 = 1.5$), rates 1.01–1.06 vs. predicted $O(\lambda_2)$.

Remark 7.3 (the remaining two acceptance limits). (iii) *The principal series returns to ordinary AdS_2 : the radial operator contracts to the Liouville operator with the Schwarzian generalised eigenfunctions and eigenvalues (Theorem 5.4, checks (R2)), and the density of states is Proposition 7.1.* (iv) *The strange series disappears, with mechanism and rates: Proposition 5.14 (checks (R7)) — at fixed label the eigenvalue merges into the band edge at distance $\approx a^2 \lambda_2^2/2$; in edge units the tower survives as the ordinary discrete series of that edge while its mirror is expelled like $4/\lambda^2$; the alternating radial profile $(-1)^k q^{2ak}$ has no continuum limit as a function (Proposition 6.2(c)). The label range does not collapse and unitarity does not fail; the disappearance is spectral merging plus expulsion — this is the “rate and manner” record requested by issue #9. The classical $SL(2, \mathbb{R}) \rtimes \mathbb{Z}_2$ matrix-element computation flagged by SI as missing (their §2.3 footnote) is not needed for any of these limits and remains open in the literature.*

7.2 The verification map

Numerical verification in this project runs alongside every claim rather than as a separate phase; this subsection is the consolidated record. All scripts are pure `mpmath` (software arbitrary precision), so every committed output in `results/` reproduces *bit-for-bit* (established under issue #20, verified across two independent machines under issue's #22 CI and the session-3 audit); each script header states the claims tested, the issue, and the run command.

script (src/)	issue	checks	what is verified
<code>check_conventions</code>	#3	(C1)–(C5), 63	dictionaries SI/GKK/BINN, series eigenvalues, T
<code>check_claim_kernels</code>	#4, #18	(K1)–(K5)	kernels, reflection identity, limit ratios, bound
<code>check_normaliser_grading</code>	#15	(G1)–(G7), 72	Π , factorisation, GKK grading/fusion blocks
<code>check_strange_coefficients</code>	#6	(S1)–(S5)	radial closed forms, Wronskian, connection
<code>check_radial_casimir</code>	#7	(R0)–(R7), 45	slice reduction, classical limit, limit circle, grids
<code>check_principal_bilocals</code>	#27	(P1)–(P5)	complex-weight kernels, transition, exact marginal line
<code>check_q1_limits</code>	#9	(L1)–(L2)	density of states, Bessel- K wavefunctions
<code>check_tex_integrity</code>	#22	(T1)–(T7)	labels/refs/cites/environments/ phrases/paths

Worst deviations among passing checks are stated inline at each claim; representative scales: 10^{-60} for exact identities at 60-digit arithmetic, 10^{-30} – 10^{-40} for closed-form checks at 40 digits, and measured-rate comparisons (never absolute tolerances) for all $q \rightarrow 1$ limits.

Failures ledger (equal prominence, CONTRIBUTING.md §5; details in LOG.md at the referenced issues). Checks or analyses that failed at first attempt, by type: (i) *an analysis error caught before commitment*: the first deficiency-index analysis of J_- concluded limit point; corrected by [15] and the measured slopes (Remark 5.8); also a sign slip in the Li_2 asymptotics first suggested a divergent $\text{AdS} \leftrightarrow \text{centre}$ amplitude, corrected by direct computation (Remark 1.10). (ii) *The recurring tolerance trap*: absolute tolerances against quantities growing like q^{-2m} or decaying like $q^{2\ell k}$ failed in (C1) (float64), the reflection check of (K), (S2), (G4), (G7), and the (R1) wall test; all were re-based on relative deviations at predicted correction scales. (iii) *Implementation*: a truncated $(a; q)_\infty$ tail in (K1); an exponentially unstable upward recursion in (R6); two test-code bugs in (R1). (iv) *Checks weaker than the mathematics*: (P5) hunted for an asymptotic drift where the crossing is exact; (P2) asserted $O(\lambda_2)$ residuals where the symmetric combination cancels to $O(\lambda_2^2)$; (R2) compared in a gauge with an $O(\lambda)$ artifact. No committed mathematical claim was falsified by a check; every failure was in a check design, an implementation, or a pre-commitment analysis draft, and each fix is recorded.

Issue #10's task map: (1) the radial q -difference equation solved numerically with spectra against analytic predictions — (R5)/(R6); (2) orthogonality/Plancherel where known — (C5), (L1), and the GKK theory cited (not re-derived); *no* orthogonality exists for the strange family on the DSSYK space (gap #14, Remark 4.5); (3) $q \rightarrow 1$ numerics including the strange disappearance — (R2), (R7), (L1)–(L2); (4) matrix coefficients against direct evaluation of the representations — (C1)–(C4), (G6)–(G7), (S1)–(S5); (5) the “scan across the AdS/dS boundary”: by Theorem 6.5(iv) no boundary is crossed by the label; the scans that exist are the extension-grid scans (R6) and the transition scans (P2)–(P5), which locate the one genuine crossing — in the bilocal weight, at the exact midpoint.

7.3 Cross-checks against independent literature

Remark 7.4 (q -Askey deformations: the strange range appears as deformed bound states). [DERIVED] (transcription of [18] §2, labels `eq:dissrete cond 1-eq:bounds discrete levels` of the arXiv source, plus one line of arithmetic.) The q -Askey-deformed chord Hamiltonians of [18] (Al-Salam–Chihara through Askey–Wilson; precisely the deformation direction SI’s “AW-DSSYK” outlook anticipates, and the polynomial-side counterparts of GKK’s half-line sector operators) acquire, for a deformation parameter $|\alpha| > 1$, a *finite* tower of bound states $E(\theta_l) = \frac{2}{\sqrt{1-q^2}} \cosh \theta_l$, $\theta_l = \log \alpha - 2\lambda l$, $0 \leq l \leq \lfloor \log |\alpha| / 2\lambda \rfloor$, with $\frac{\sqrt{1-q^2}}{2} E \in (1, \mu(\alpha)]$ for $\alpha > 1$ and $\in [-\mu(|\alpha|), -1)$ for $\alpha < -1$. In our variables these are $\tilde{\Omega}$ -values $+\mu(|\alpha|e^{-2\lambda l})$, resp. $-\mu(|\alpha|e^{-2\lambda l})$ (for $\alpha < 0$, $\log \alpha = \log |\alpha| + i\pi$ makes $\cosh \theta_l$ negative — the same half-period shift as everywhere in this project): *the $\alpha < -1$ towers sit exactly in the strange range $\tilde{\Omega} < -1$, log-periodically spaced, truncated at the $|y| > 1$ -type window — the finite-tower analogue of our wedge grids (30) and of GKK’s half-line sector spectra. [18] interpret them as ETW-brane/Morse-potential*

bound states, non-perturbative in G_N , and do not connect them to the strange series (consistent with the issue-#2 finding that no citing paper engages it); the identification of their $\alpha < -1$ family with strange-series Casimir values appears to be new to this project. *Verdict: agree* — independent appearance of beyond-the-band bound states at strange values, with the negative-parameter/alternating structure matching Lemma 1.4’s $t < 0$ weights; their Morse potential is the two-exponential completion of our one-exponential well (Remark 5.2).

Remark 7.5 (sine-dilaton gravity: the curvature flip sits at the band centre). [DERIVED] (transcription of [16] §2, §5.2, with the convention map.) Sine-dilaton conventions have $E = -\cos \theta_{\text{BMP}}/(2|\log q|)$ with ground state at $\theta_{\text{BMP}} = 0$, so $\theta_{\text{BMP}} = \pi - \theta$ in our Convention 2.1. Their classical solutions have $F(r) = -2\cos r + 2\cos \theta_{\text{BMP}}$, horizon at $r = \theta_{\text{BMP}}$, and Ricci scalar $R = -2\cos(2|\log q|\Phi)$; at the horizon, $R = -2\cos \theta_{\text{BMP}} = +2\cos \theta = \sqrt{1 - q^2} E$. Hence: the horizon curvature is *negative (AdS-sign) at the ground-state edge $\theta = \pi$* — agreeing with Definition 1.2 — *positive (dS-sign) at the opposite edge*, supporting the edge-dS reading their follow-up [3] makes precise, and *flips sign exactly at the band centre $\theta = \pi/2$* , where their static-patch “pode” discussion also sits. *Verdict: agree*, threefold: (i) the AdS edge assignment matches; (ii) their effective-curvature criterion is literally $\text{sgn } E$ — the orientation datum of Definition 6.1(II), consistent with Proposition 6.2(a) (in the bulk dual the curvature is an energy-orientation datum, not a representation datum); (iii) their curvature-flip locus coincides with the exact transition locus of Theorem 6.8(iv) and with the centre-dS point — three independently derived structures at $\theta = \pi/2$.

Remark 7.6 (centre-dS). [ARGUED] Under the centre-dS identification [5], the strange bilocal $\text{AdS} \leftrightarrow \text{dS}$ amplitude vanishes like $\exp(-3\pi^2/4|\ln q^2|)$ (Remark 1.10, measured rates $-4.97/-7.07/-7.22$ vs. -7.40), so Claim 1.7 fails there — but the transition locus of Theorem 6.8 sits exactly at the centre-dS point: in the weight parameter, edge-connectivity switches on precisely at the weight whose spectral image is their dS locus. *Verdict: disagree on the claim under test (as SI’s own footnote anticipates), with the new structural coincidence recorded.*

Remark 7.7 (chord von Neumann algebra, type II_1). [ARGUED] [17] establishes the (zero-particle) chord algebra as a type II_1 factor. Consistency on shared quantities: our chord-sector spectral result (Proposition 3.6: purely absolutely continuous band, no point spectrum) is what the trace representation of a II_1 factor generated by T requires — no normalisable transfer-matrix bound states in the chord sector; and our tower spectra live on the $\epsilon = +1$ wedge, *outside* the chord sector, so their absence from the II_1 analysis is consistency, not tension. The follow-up [19] computes cosmological entanglement entropy from the same algebra; it shares no computed quantity with this project beyond that consistency. *Verdict: consistent (orthogonal quantities).*

Remark 7.8 (quantum symmetry in the chord algebra). [ARGUED] [20] construct $U_q(\mathfrak{su}(1,1))$ generators inside the chord algebra and show the *one-particle* chord space decomposes into *positive discrete series* D_ℓ^+ only. This is exactly what the classification of Convention 2.5 forces — a one-sided K -spectrum admits no strange constituents (two-sided is automatic for the strange family) — and it corroborates, from an independent construction, this project’s structural finding that strange constituents require the two-sided/normaliser structures (the fusion of Theorem 1.16), not the one-sided matter modules. Their quantum-disk realisation concerns the Cartan-type quotient that Definition 3.1 records as *distinct* from the DSSYK coset (SI §5 verbatim), and their $U_q(\mathfrak{sl}(2, \mathbb{R}))$ -vs- $U_q(\mathfrak{su}(1,1))$ discussion parallels SI’s Cayley-transform caveat (quoted in Section 5 sources). They do not engage the strange series (issue #2 grep). *Verdict: consistent; no shared strange-series quantity exists to compare.*

Remark 7.9 (summary of the cross-checks). No contradiction with any independent source was found; two genuine convergences emerged: the strange-range bound-state towers (independently

present in [18] as $\alpha < -1$ deformations, unidentified there), and the band-centre locus, at which three independent structures meet — the sine-dilaton curvature flip, the centre-dS proposal, and this project’s exact edge-connectivity transition. All comparisons passed through the translation table (Convention 2.8); the one convention trap encountered ($\theta_{\text{BMP}} = \pi - \theta$) is stated where used.

8 Conclusion and open gaps

8.1 Verdict

The question of [1] — do the strange series representations genuinely interpolate between the AdS_2 and dS_2 regions of the DSSYK bulk dual? — splits, on formalisation (Section 1.3), into the amplitude claim C-amp and the geometric claim C-geo, and the verdict (Theorem 6.5) is:

- *C-amp holds*, sharpened: at finite q the strange kernel is the discrete kernel with one spectral argument reflected through the band centre (Lemma 1.5, an exact identity, not a limit), so the mixed-edge strange amplitude *equals* the same-edge discrete amplitude; the claim is conditional only on the edge-dS identification, fails under centre-dS, and its selectivity refines, on the full complex-weight bilocal family, to an exact transition at the band-centre weight (Theorem 6.8).
- *C-geo is refuted*: every intrinsic criterion is normaliser-covariant and cannot separate the strange family from its fused discrete partners (Theorems 1.21, 6.5); the orientation separates them but places the *strange* towers at the AdS edge; no quantity crosses as the label varies. The $\text{AdS} \leftrightarrow \text{dS}$ bridge the framework supports is the normaliser grading Π itself, of which the strange series is one half of an orbit.

This is outcome 3 plus outcome 2 of PROBLEM.md §6: a negative result with an explicit obstruction, together with sharp structures — the radial-part theory of Section 5 and the falsifiable residue Conjecture 6.6 — that make the remaining question precise.

8.2 What stands independently of the verdict

For a reader of [1], the following are usable regardless of any AdS/dS position: the factorisation $\mathcal{S}^\ell = \mathcal{O}^\ell \Pi$ through the chord parity and its essential uniqueness (Proposition 1.14, Lemma 1.18); the fusion transcription with the label map $a = \ell - \frac{1}{2}$ (Theorem 1.16); the slice reduction of SI (4.25) to two radial operators and the identification of the wedge operator with Koelink’s case-(1) family — deficiency indices $(1, 1)$, per-extension spectra on two bilateral q^2 -grids with the root product rule (31) and the fusion-realising Π -invariant extensions (Section 5.4); the closed-form radial solutions and their $q \rightarrow 1$ confluences to Bessel I/K through the connection formula (Theorem 4.2, Proposition 7.2); the exact marginal-line identity and midpoint crossing (Theorem 6.8); and the identification of the $\alpha < -1$ q -Askey bound-state towers of [18] with strange-series Casimir values (Remark 7.4). Two suspected misprints in [1] v2 are recorded in lit/notes/2512.10101.md §7 (the Γ_{q^2} shift in (4.19), twice).

8.3 Open gaps, with the technique each plausibly needs

Ordered by estimated proximity.

1. *Extension selection* (issue #26, [GAP: ISSUE #26]). Transport GKK’s affiliation-fixed domain condition ($u_{-k} = q^{k/2} f(q^k)$, f differentiable at 0; their Thm. thm:Jacobioperatorforcas=q) through SI’s rescaling and match it to Koelink’s θ -parametrisation [15]. Conjecturally selects the Π -invariant fusion extension (33). A finite computation with a clear target; does not affect the verdict either way.

2. Π as a rescaled limit of $\hat{M}_-$ (issue #16, [\[GAP: ISSUE #16\]](#)). Rescale the matrix elements of GKK’s odd generator U_0^{-+} in the appropriate sector and compare with Π times a spectral phase (Lemma 1.18 fixes the target up to that phase). Same flavour as #26.
3. *Rescaled Plancherel theory* (issue #14, [\[GAP: ISSUE #14\]](#)). Either prove strong-resolvent-type convergence of the GKK sector transforms under $r \rightarrow \infty$ with control of the discrete summands, or exhibit their escape — the latter would itself locate “where the strange series goes” in the DSSYK limit. A real theorem either way; the wedge extension theory of Section 5.4 is the natural limit object.
4. *Non-spherical bilocals* (issue #13). The operators SI found “impractical”: their action must be computed from the Gauss-decomposition formulas (SI App. C) rather than from the functional calculus of $\hat{\rho}_{st}$. Hard by the authors’ own account; the transition of Theorem 6.8 says what to expect of any family that is diagonal enough.
5. *The spectral-triple metric* (SI’s announced programme). Conjecture 6.6 states what this project predicts for it: Π -covariance on intrinsic data (else Claim 1.8 reopens), and edge-locality of the towers matching Proposition 6.4.

Every gap-tagged statement in this document carries one of the three open **gap** issues #14, #16, #26 above; issue #13 is open work, not a gap in any argument made here; no other unproved step is load-bearing.

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